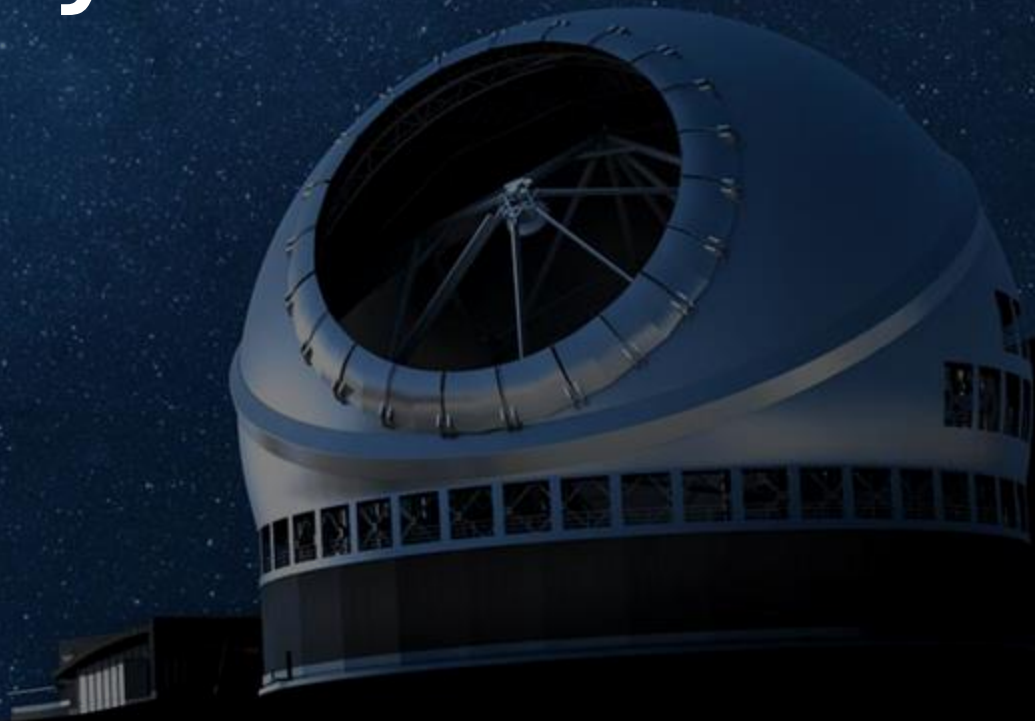


Observatory

Isabella, Kevin, Mandi,
Liliana, Hiro, Alicia



Introduction

Our group created an observatory because we were interested in space. Our original idea was a telescope but we realized we did not have enough parts, so we went with making an observatory.

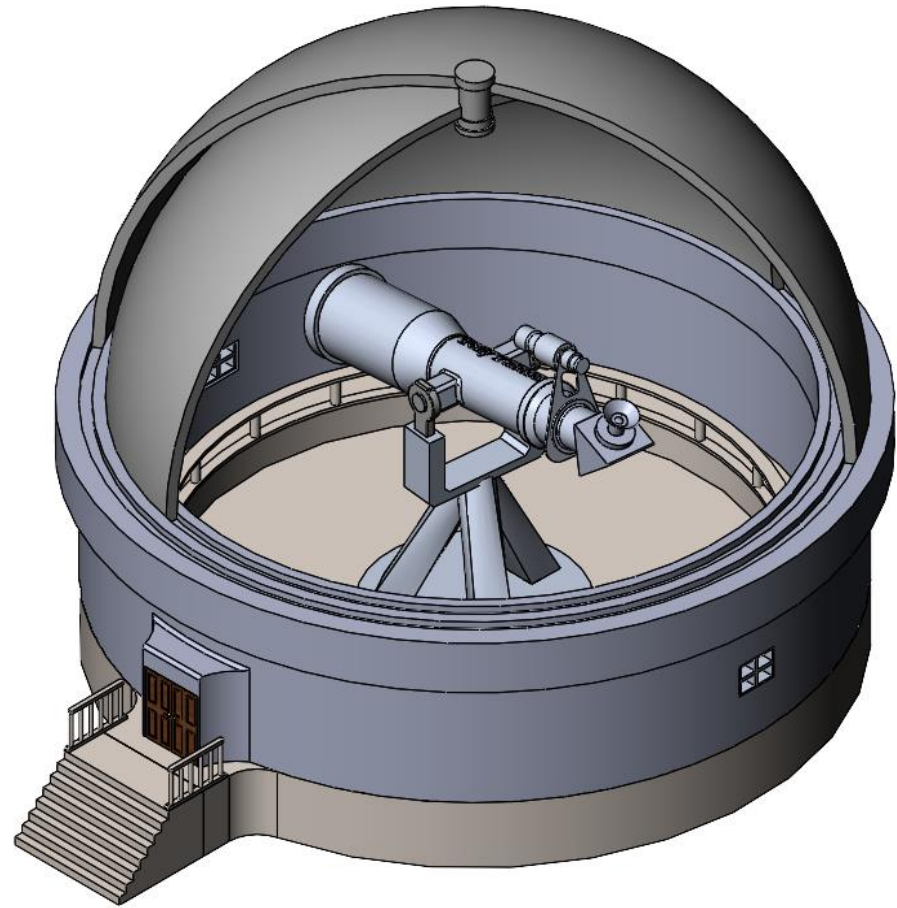
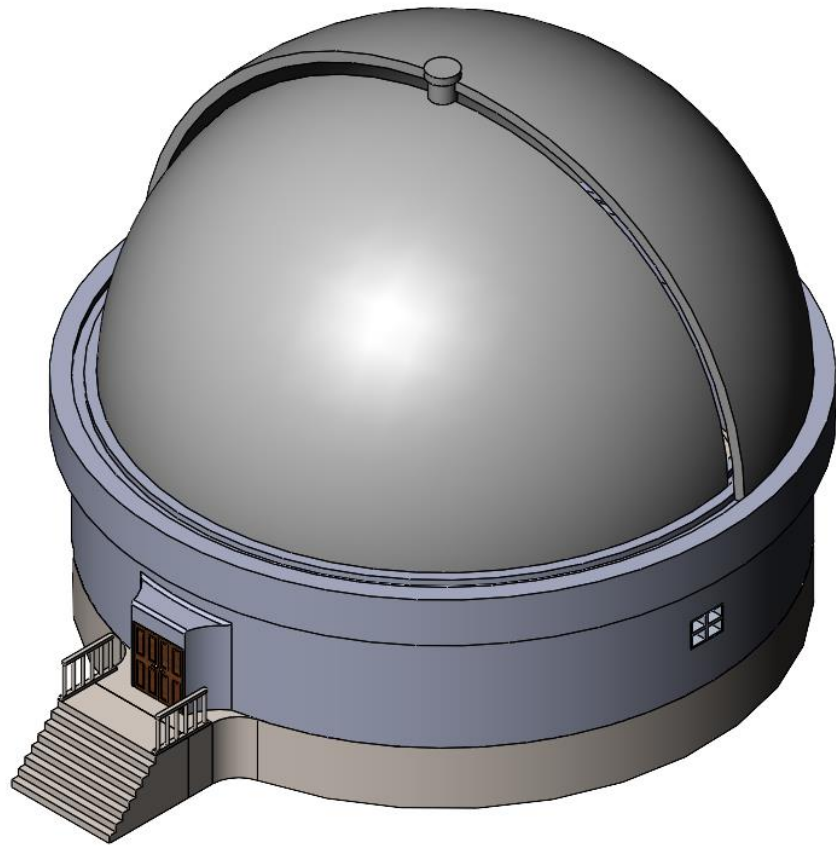
We did both a static and thermal FEA as we really wanted to see if our observatory would be able to handle the conditions that we applied to it.

We animated all of the moving parts, including the top dome, and both translational and rotational parts of the telescope, and the doors.

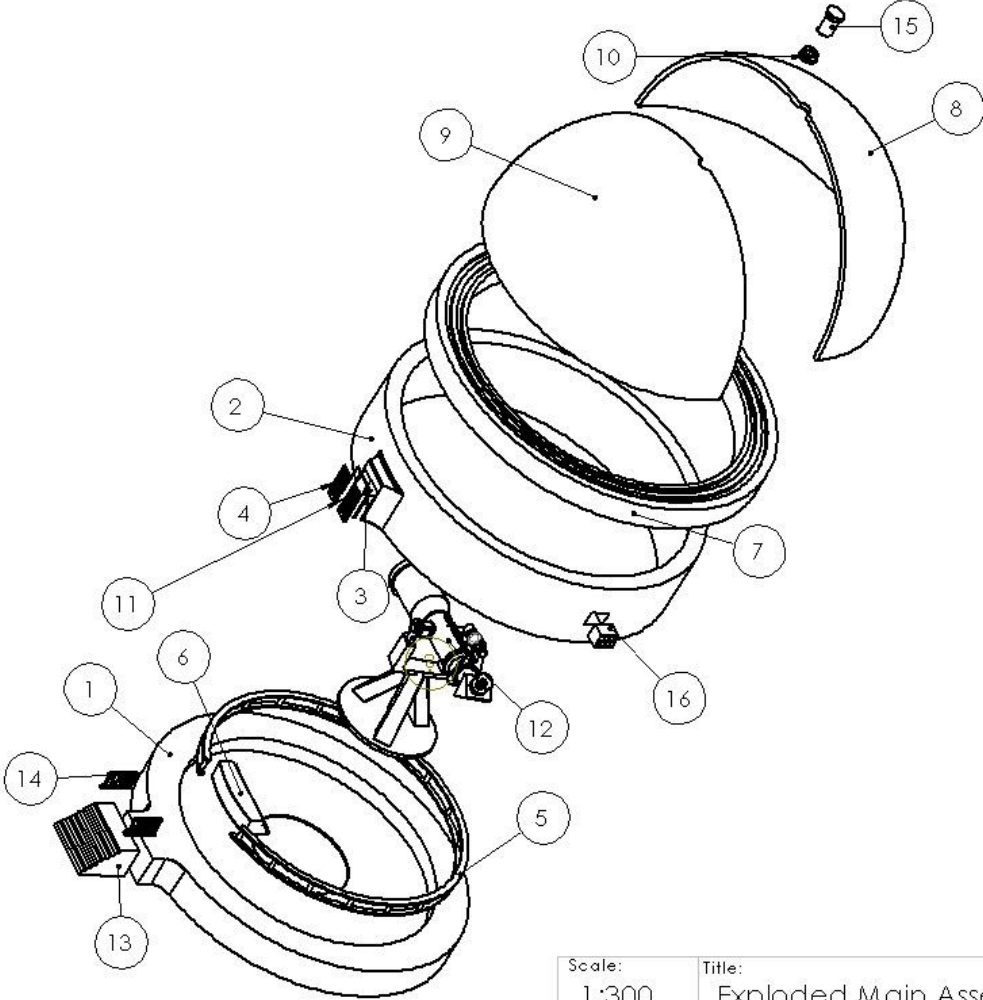
Our new feature is the 3D Sketch tool, allowing us to create a whole part with one sketch.

We used reference images and a reference model, but no tutorial videos to help us create the model.

Main Assembly



Main Assembly



ITEM NO.	PART NUMBER	QTY.
1	Floor	1
2	observatory base	1
3	Door Waist	1
4	Door	2
5	Inside Railing	1
6	Inside Stairs	1
7	dome holder final	1
8	outer dome final	1
9	inner dome final	1
10	pin cap	1
11	Door Handle	4
12	Telescope and Stand Assembly	1
13	Outside Stairs (3D sketch)	1
14	Outside Railing (Single)	2
15	pin final	1
16	Window	2

Scale:

1 :300

Units

mmgs

Title:

Exploded Main Assembly

Drawn by:

Alicia Nguyen

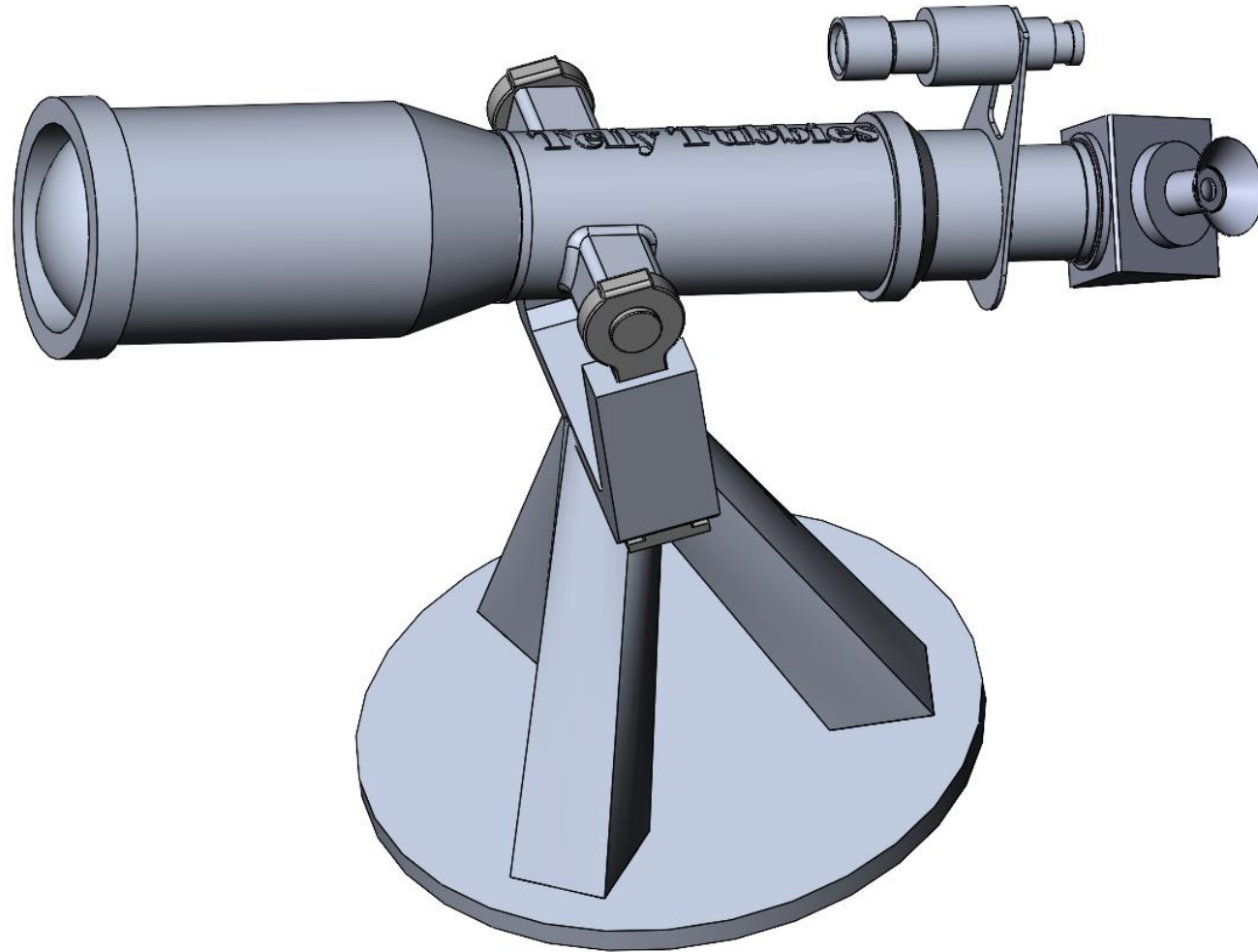
ENME 272

Fall 2025

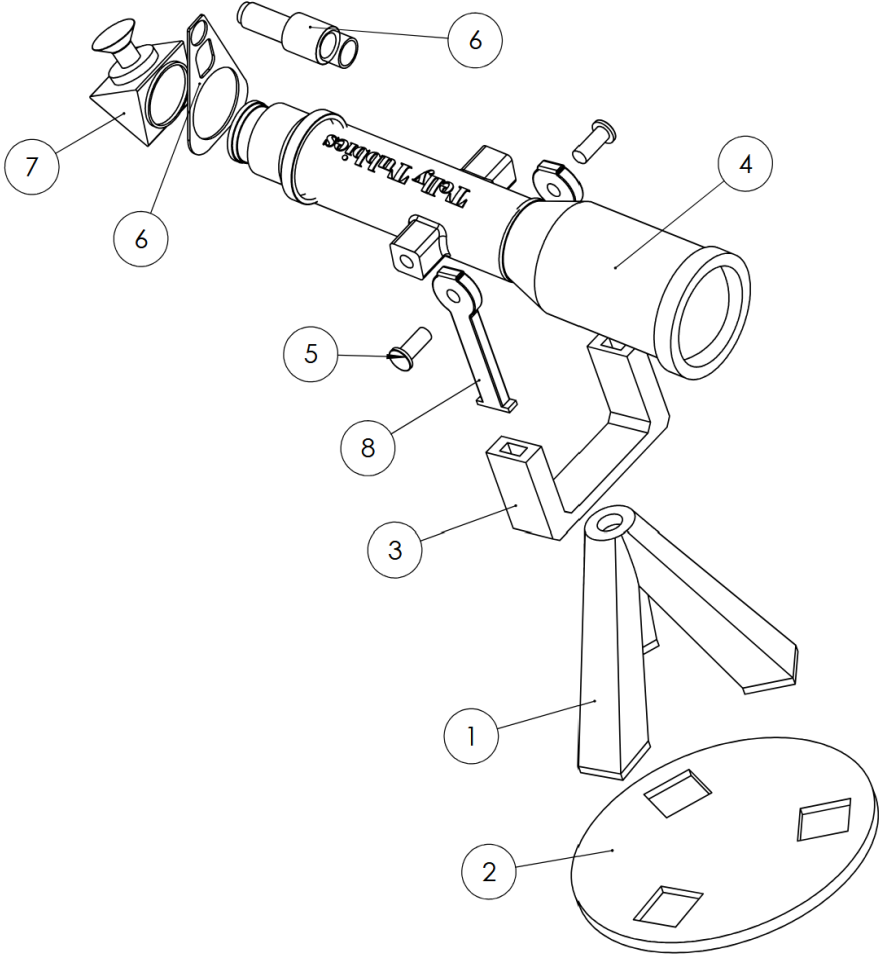
Date:

12/9/2025

Subassembly – Telescope and Stand



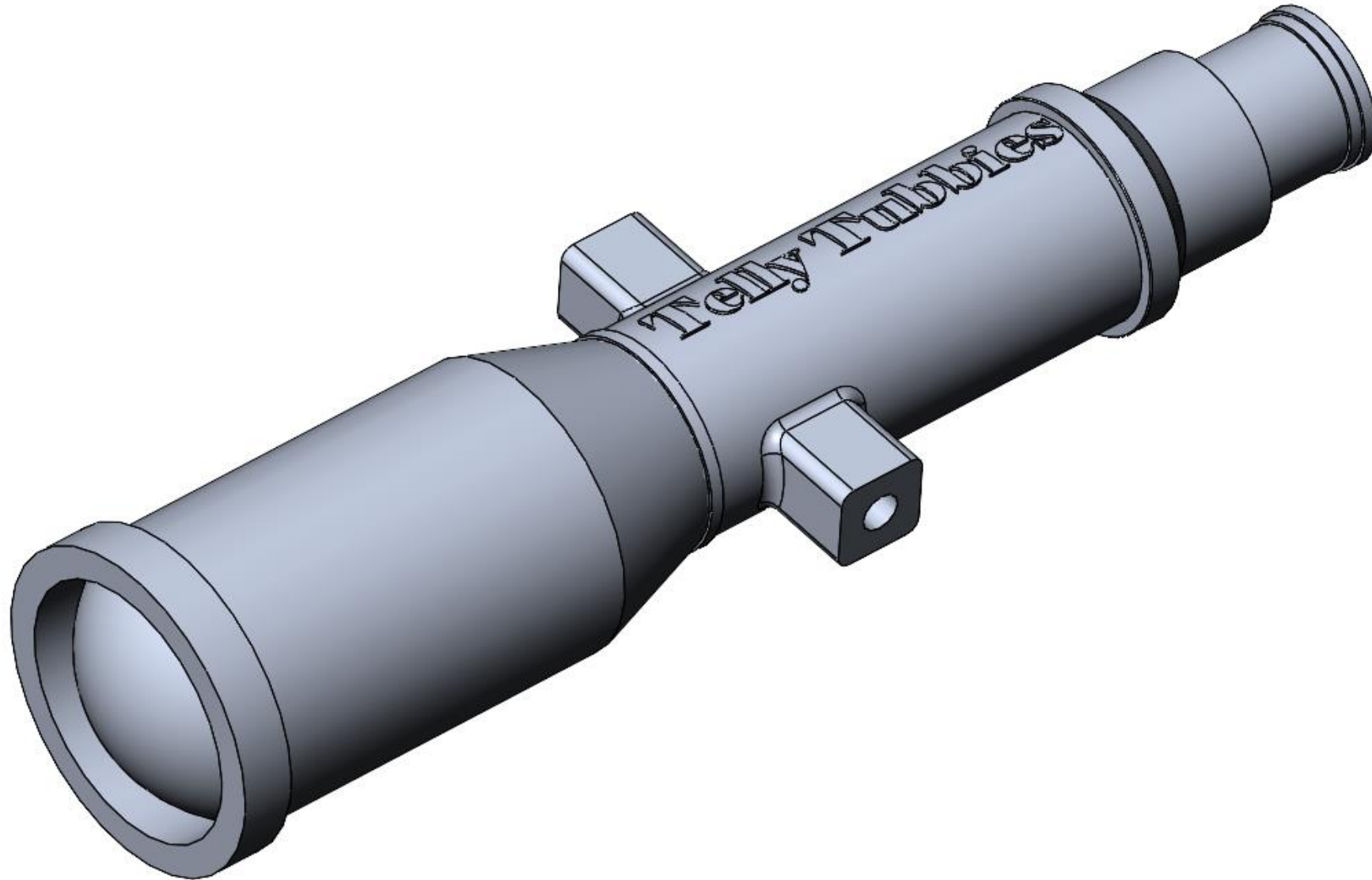
Subassembly (Telescope and Stand)



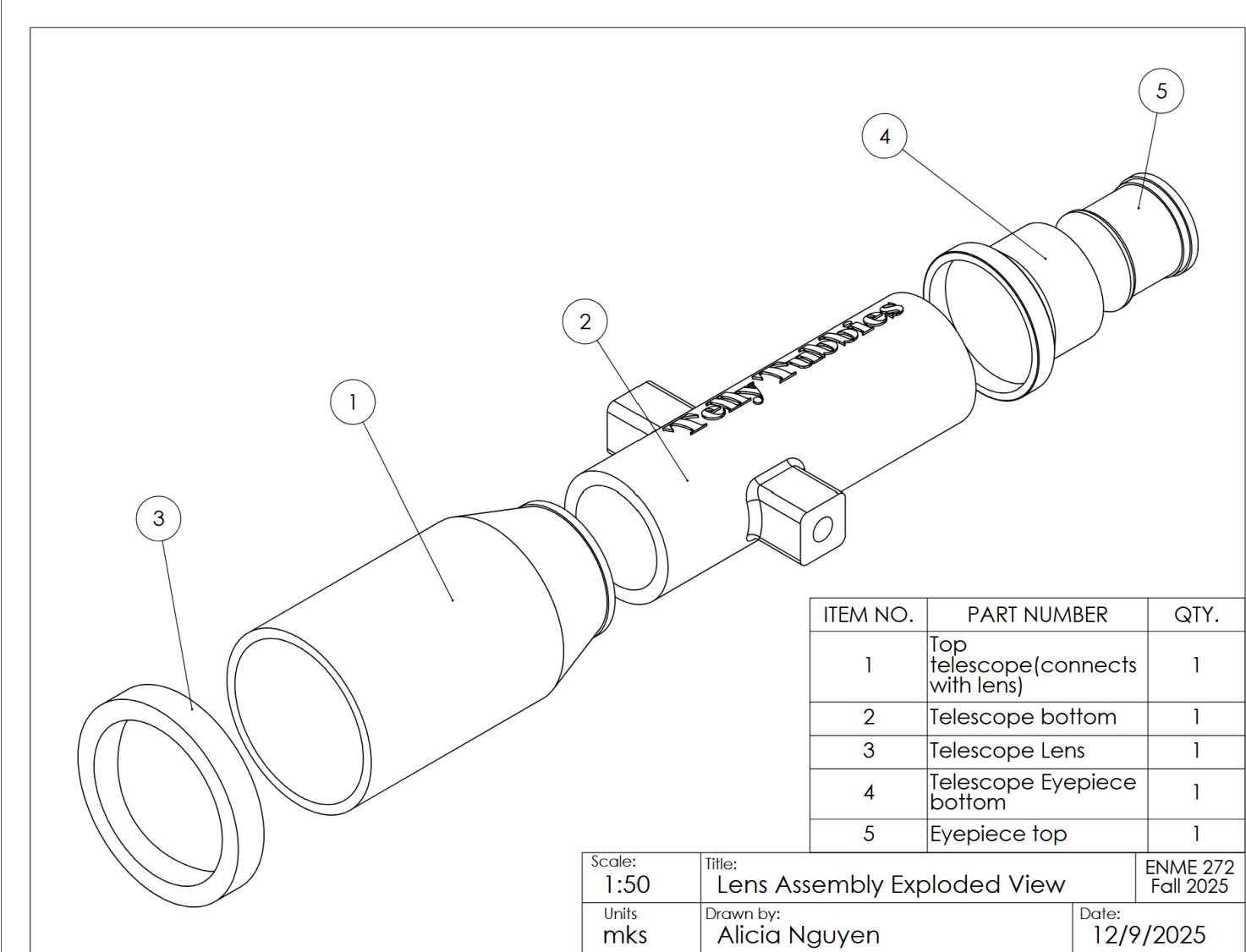
ITEM NO.	PART NUMBER	QTY.
1	Stand	1
2	Part1	1
3	Telescope holder	1
4	Telescope Assembly	1
5	Telescope Attachment Pin	2
6	assembly small scope	1
7	Watchglass	1
8	Telescope Holder Sliders	2

Scale: 1:105	Title: Telescope & Stand Exploded View	ENME 272 Fall 2025
Units mks	Drawn by: Alicia Nguyen	Date: 12/9/2025

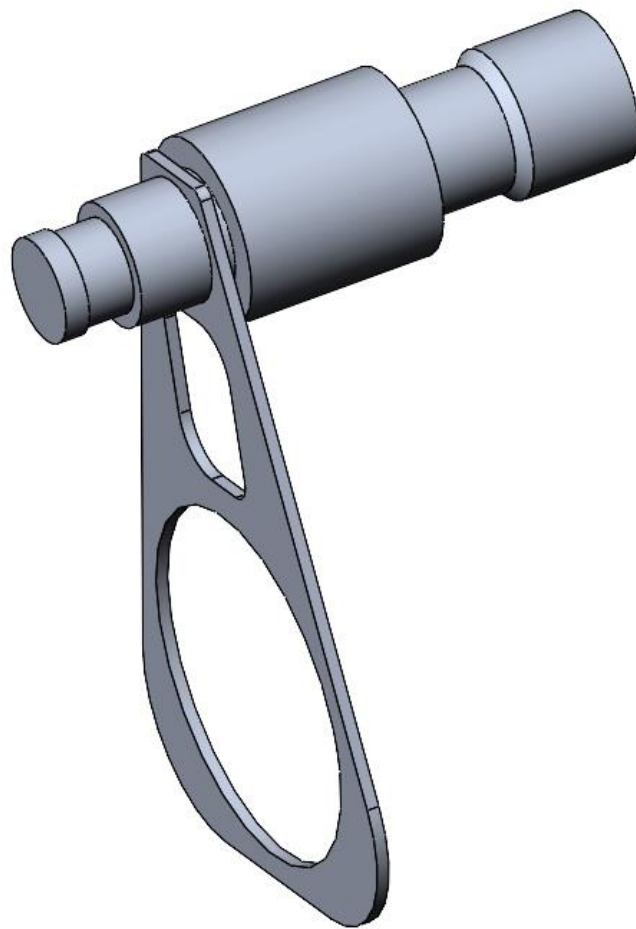
Subassembly (Lens Assembly)



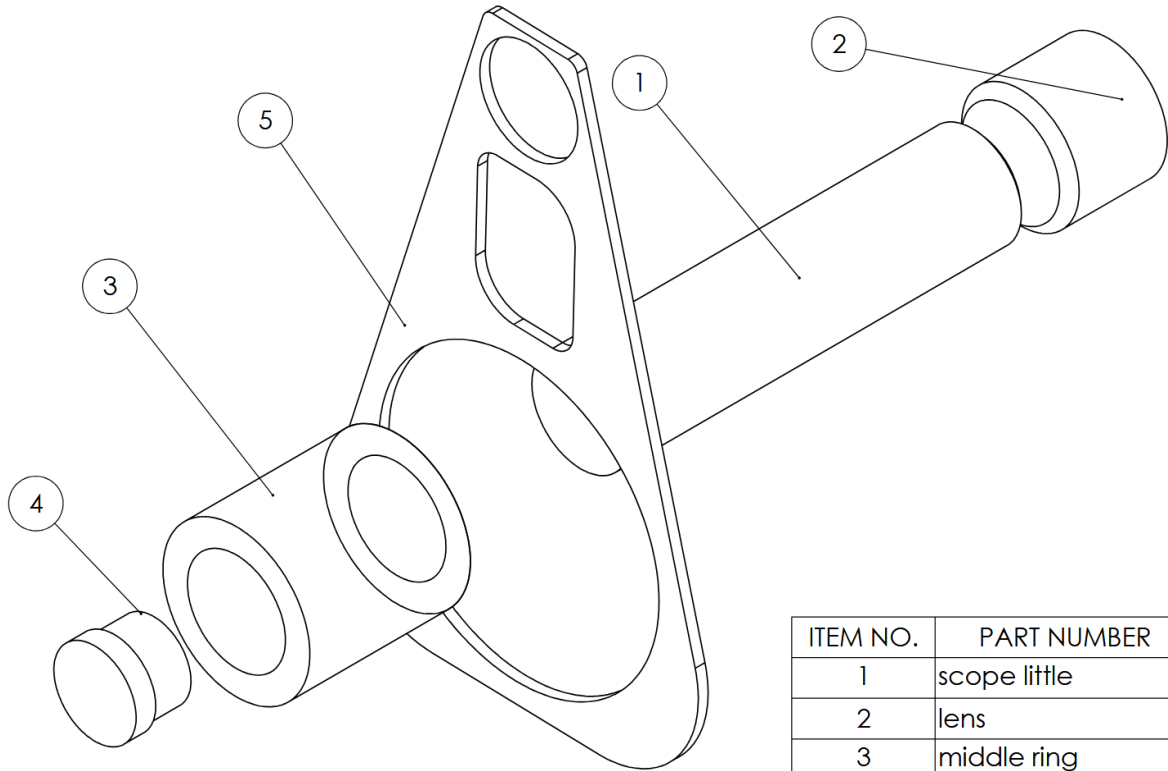
Subassembly (Lens Assembly)



Subassembly – Small Scope



Subassembly (Small Scope)

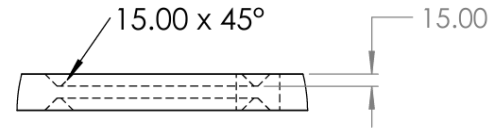
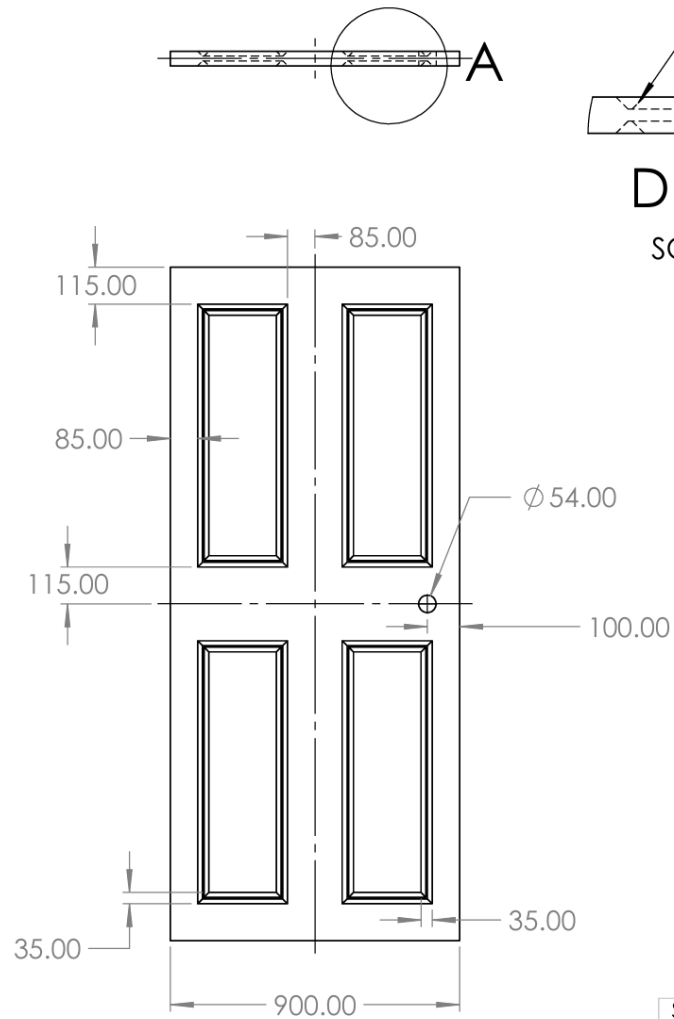


ITEM NO.	PART NUMBER	QTY.
1	scope little	1
2	lens	1
3	middle ring	1
4	end	1
5	connector	1

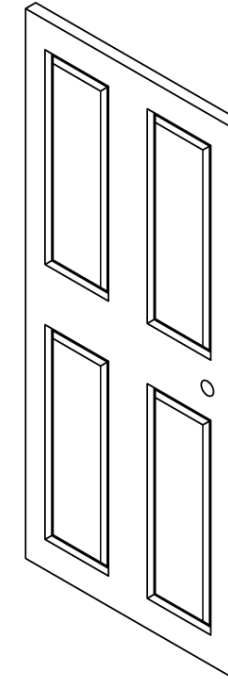
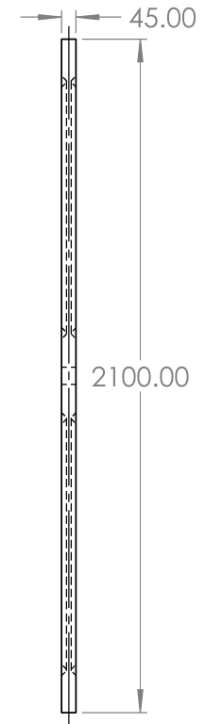
Scale: 1:20	Title: Small Scope Assembly Exploded	ENME 272 Fall 2025
Units mks	Drawn by: Alicia Nguyen	Date: 12/9/2025

Door





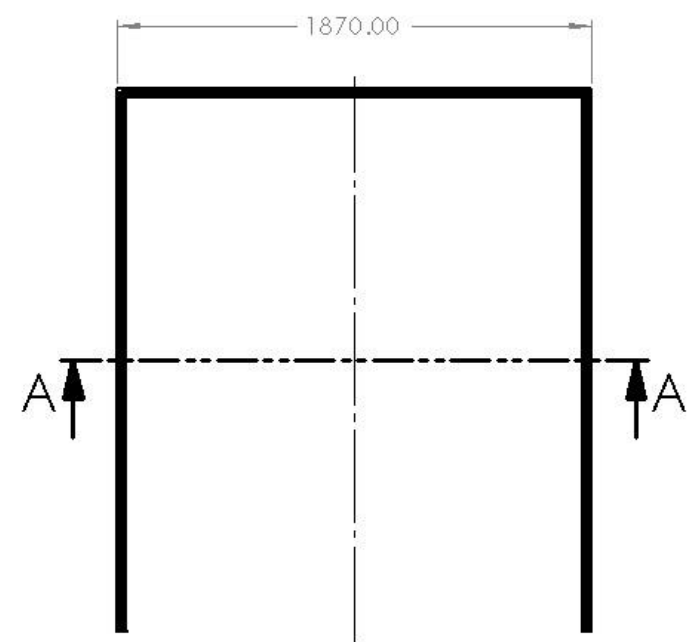
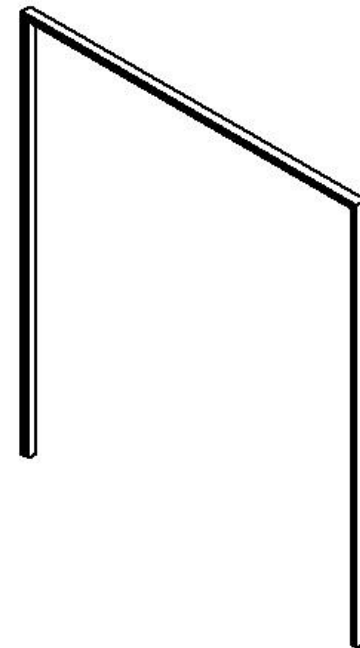
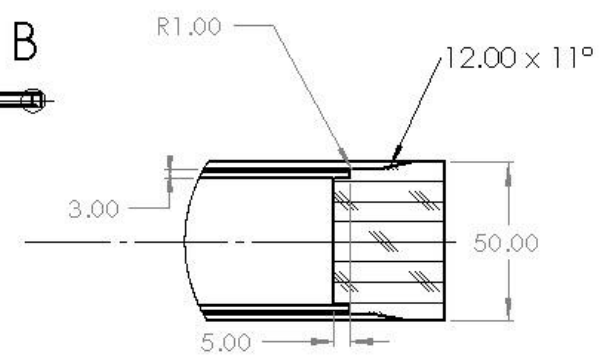
DETAIL A
SCALE 1 : 8



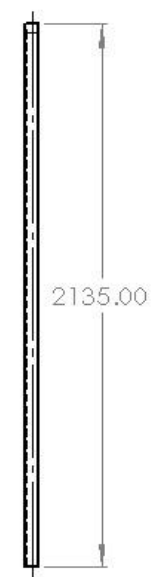
Scale: 1:20	Title: Door	ENME 272 Fall 2025
Units mmgs	Drawn by: Kevin Tish	Date: 11/18/2025

Door Waist



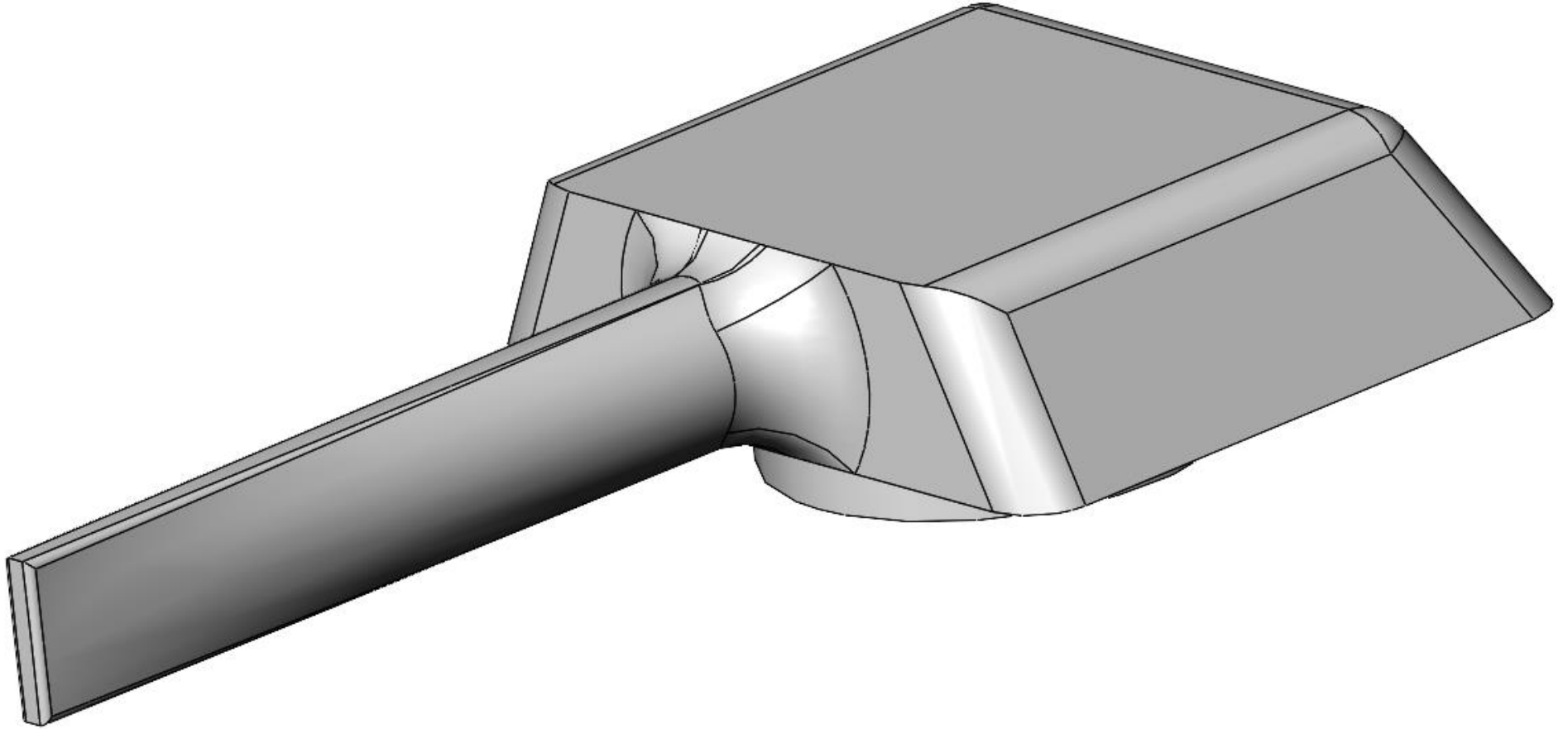


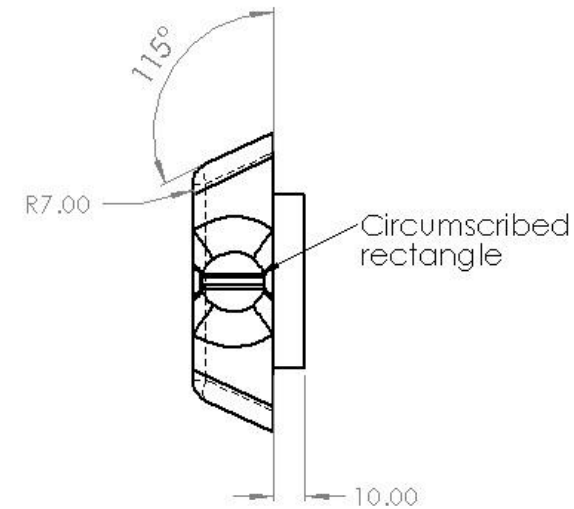
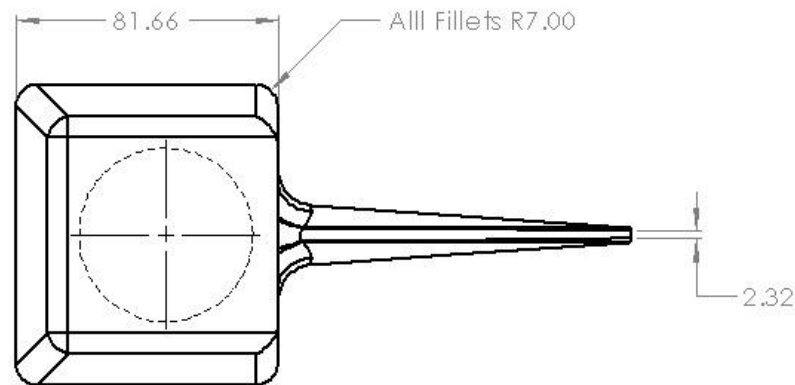
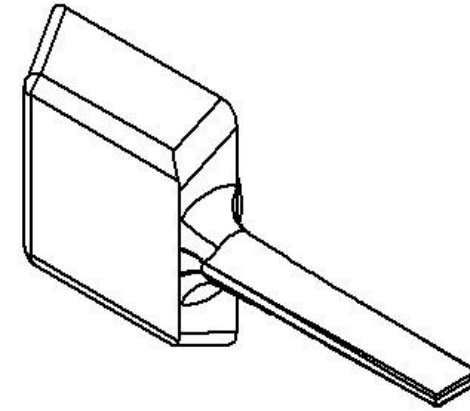
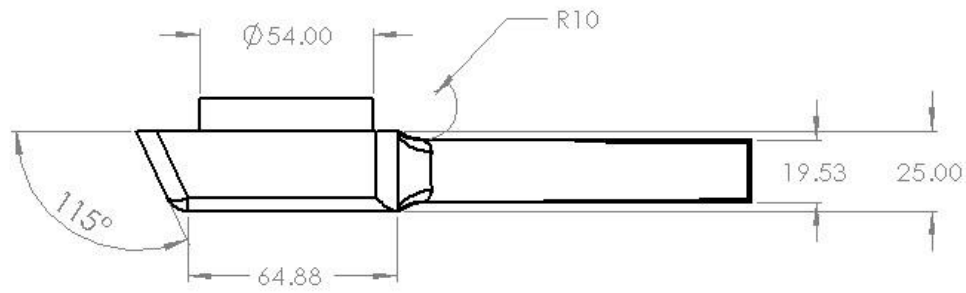
DETAIL B
SCALE 1 : 2



Scale: 1:25	Title: Door Waist	ENME 272 Fall 2025
Units mmgs	Drawn by: Kevin Tish	Date: 12/9/2025

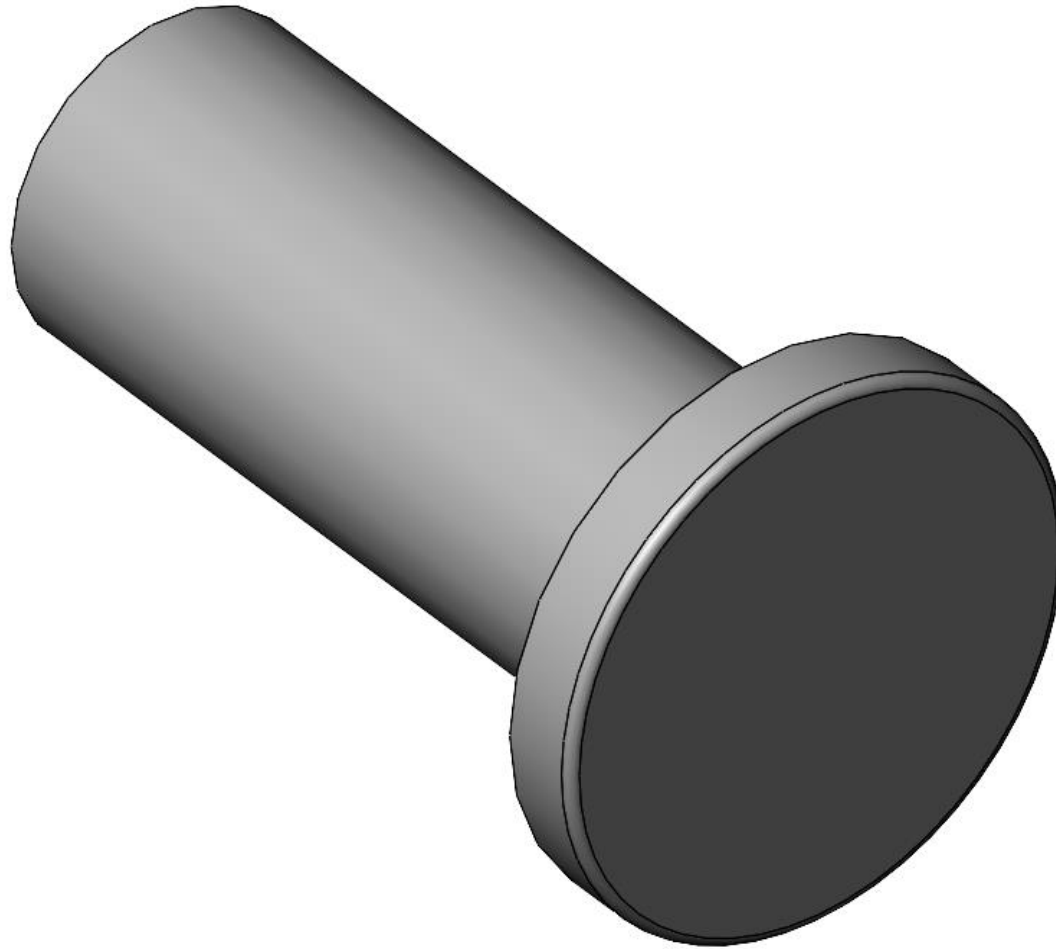
Door Handle

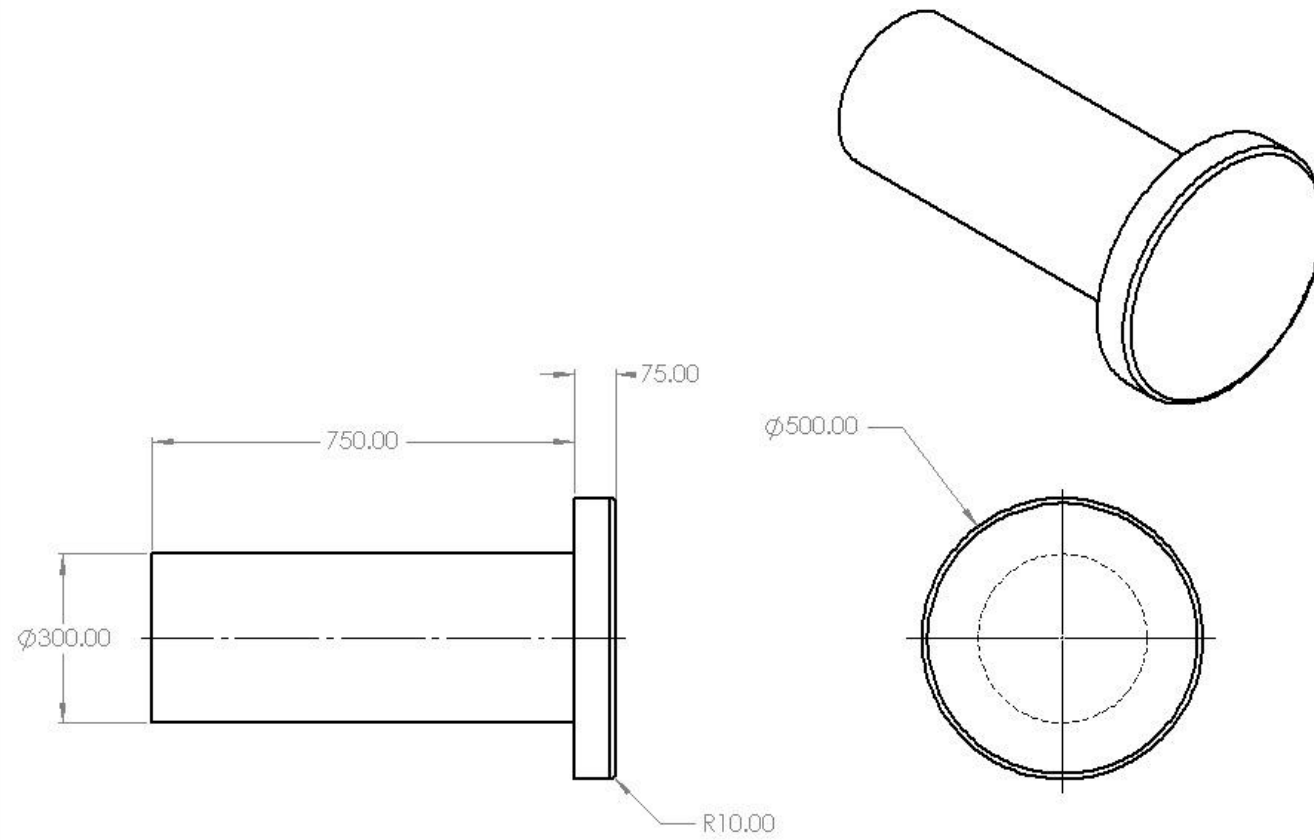




Scale:	Title:	ENME 272
1:2	Door Handle	Fall 2025
Units	Drawn by:	Date:
mmgs	Kevin Tish	12/9/2025

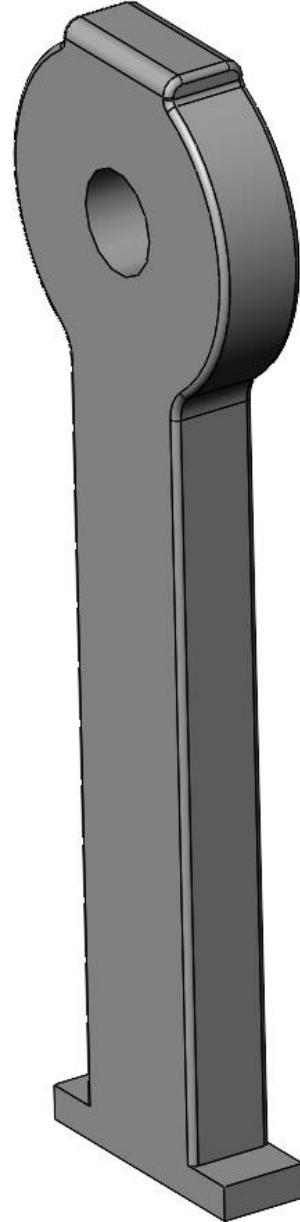
Telescope Attachment Pin

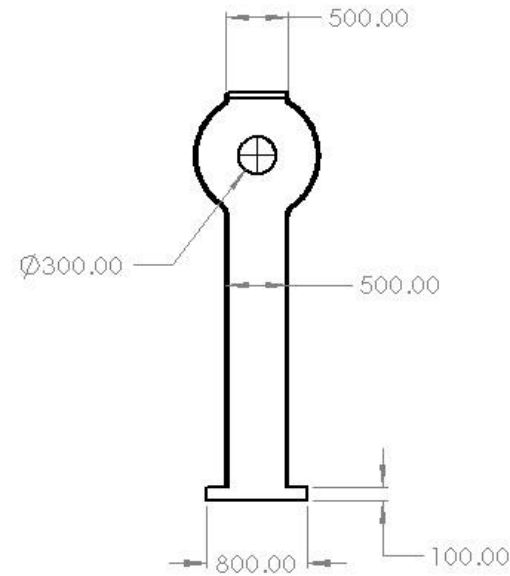
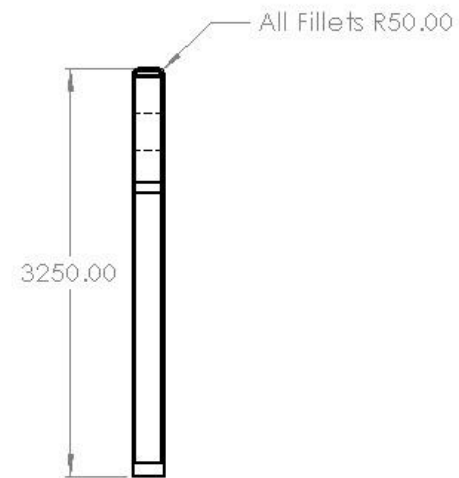
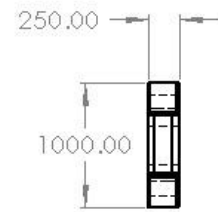




Scale: 1:10	Title: Telescope Attachment Pin	ENME 272 Fall 2025
Units mmgs	Drawn by: Kevin Tish	Date: 12/9/2025

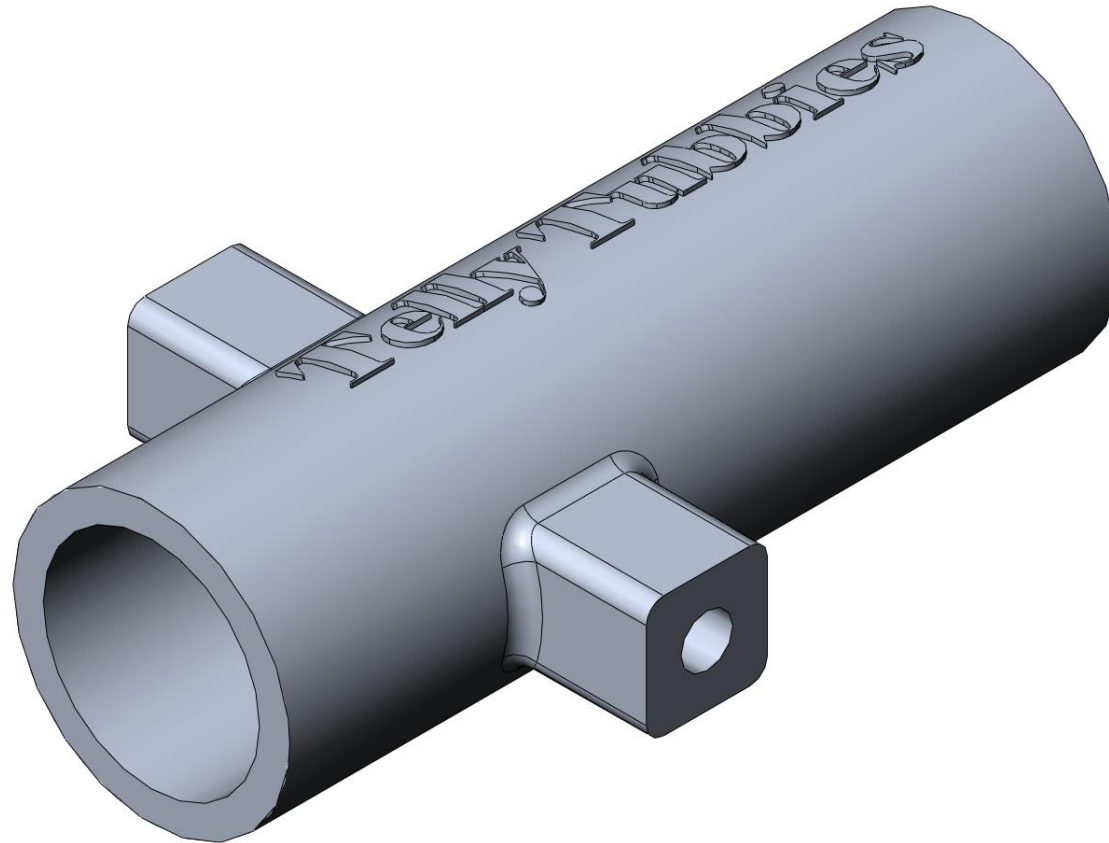
Telescope Holder Slider

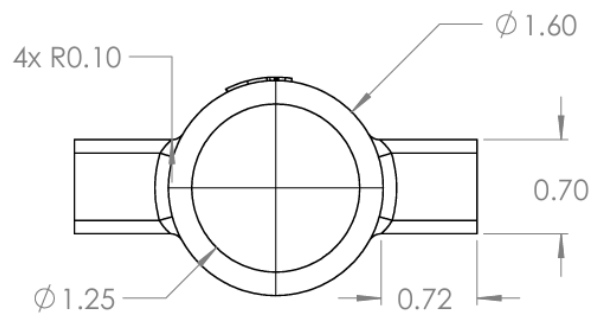
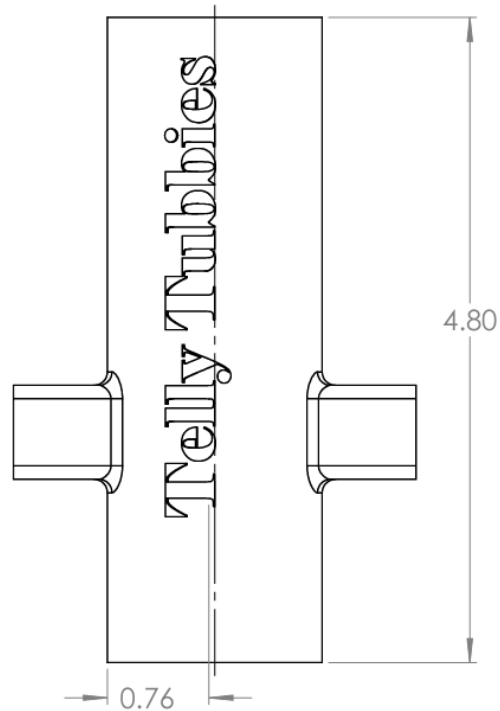




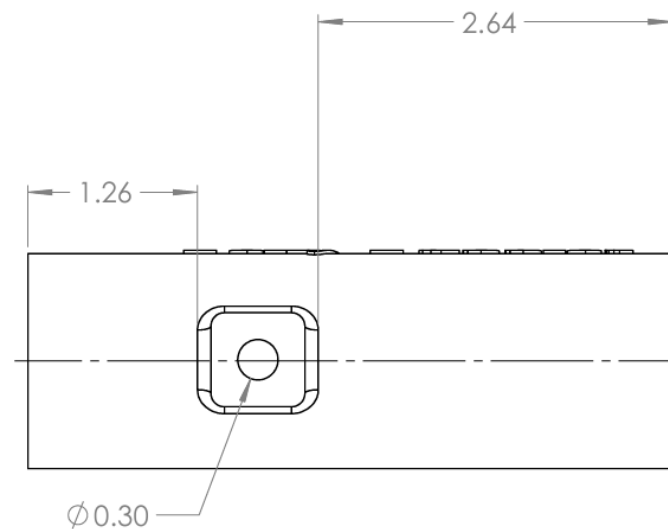
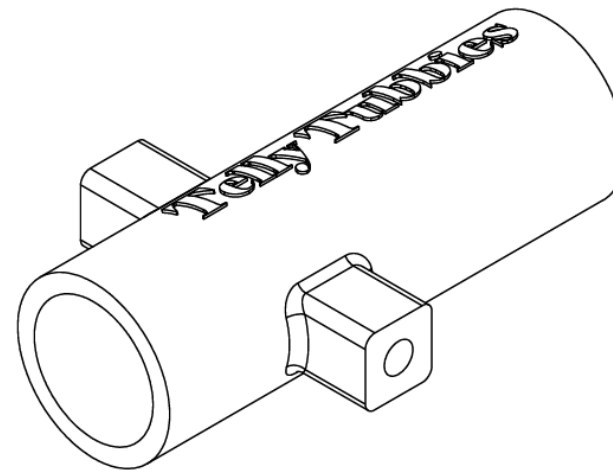
Scale: 1:50	Title: Telescope Holder Sliders	ENME 272 Fall 2025
Units mmgs	Drawn by: Kevin Tish	Date: 12/9/2025

Telescope Bottom



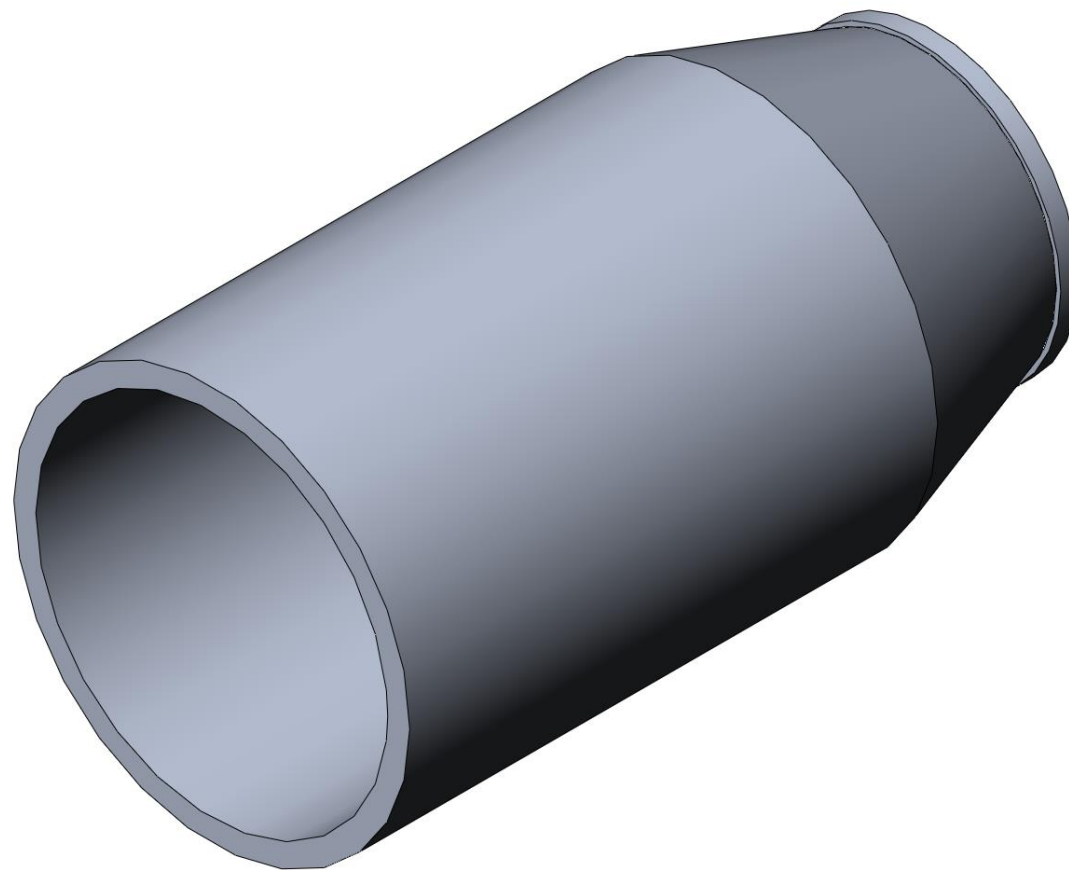


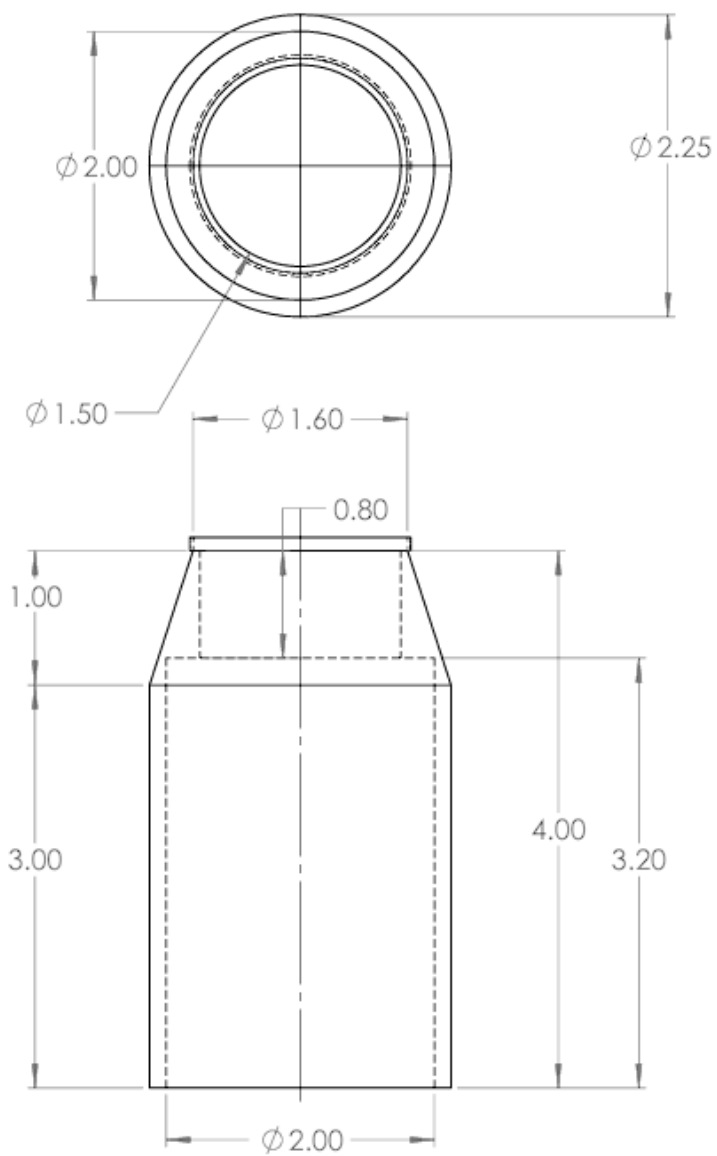
Material: Aluminum 6061



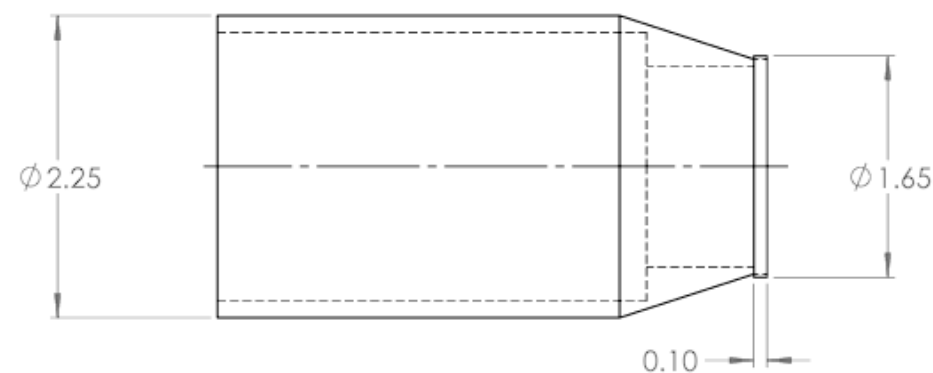
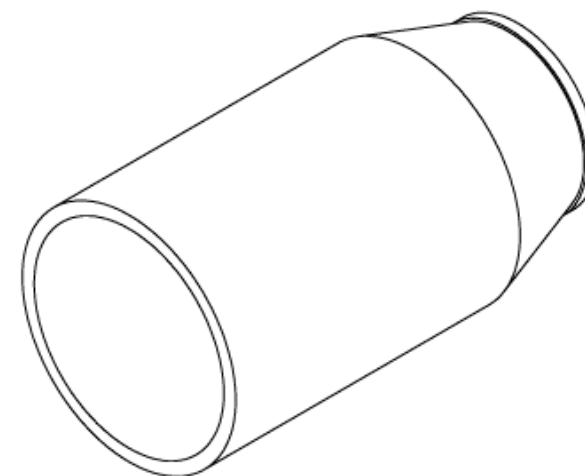
Scale: 1:50	Title: Telescope bottom	ENME 272 Fall 2025
Units Meters	Drawn by: L.Montilla	Date: 12/8/2025

Telescope top





Material: Aluminum 6061
Mass: 9213.10 kg



Scale:
1:50

Title:
Top telescope(connects with lens)

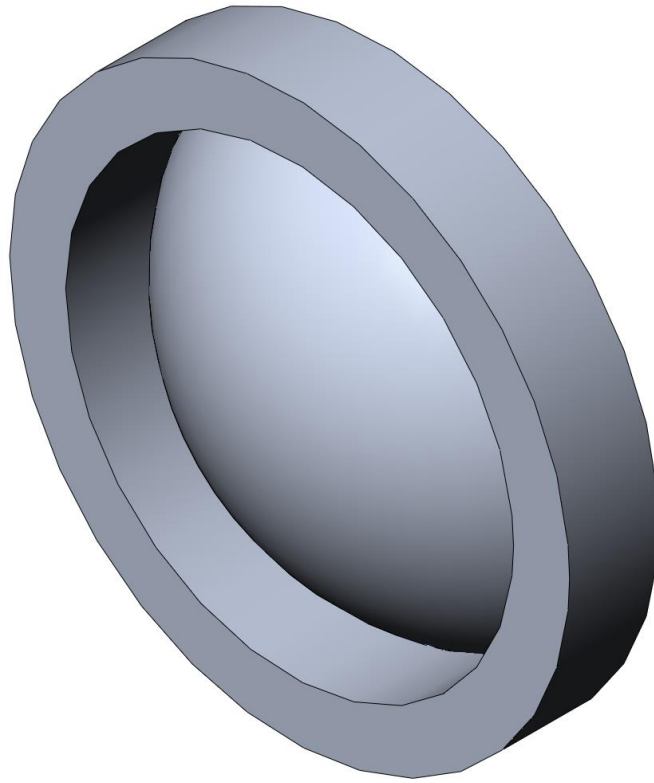
ENME 272
Fall 2025

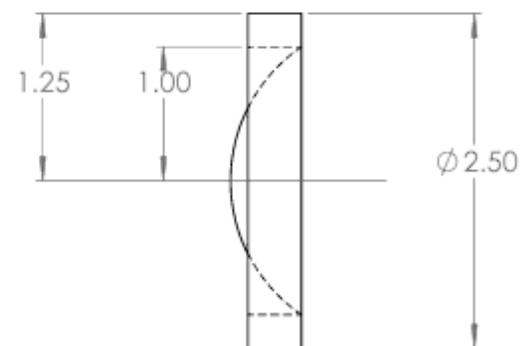
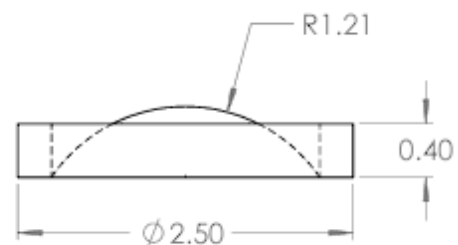
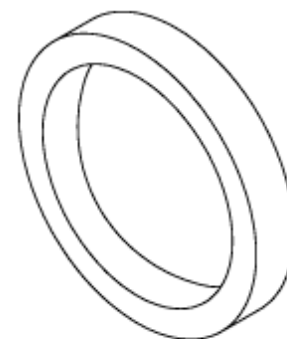
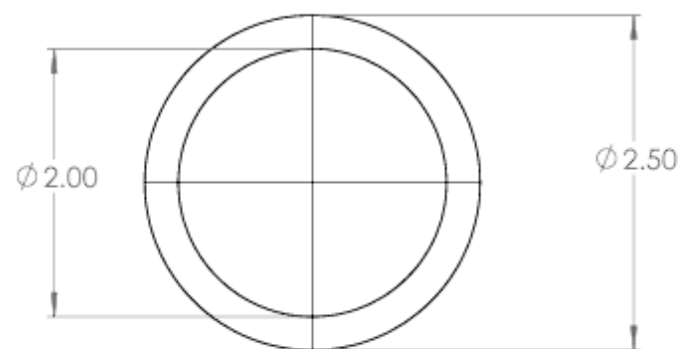
Units
Meter

Drawn by:
L.Montilla

Date:
12/7/2025

Telescope Lens

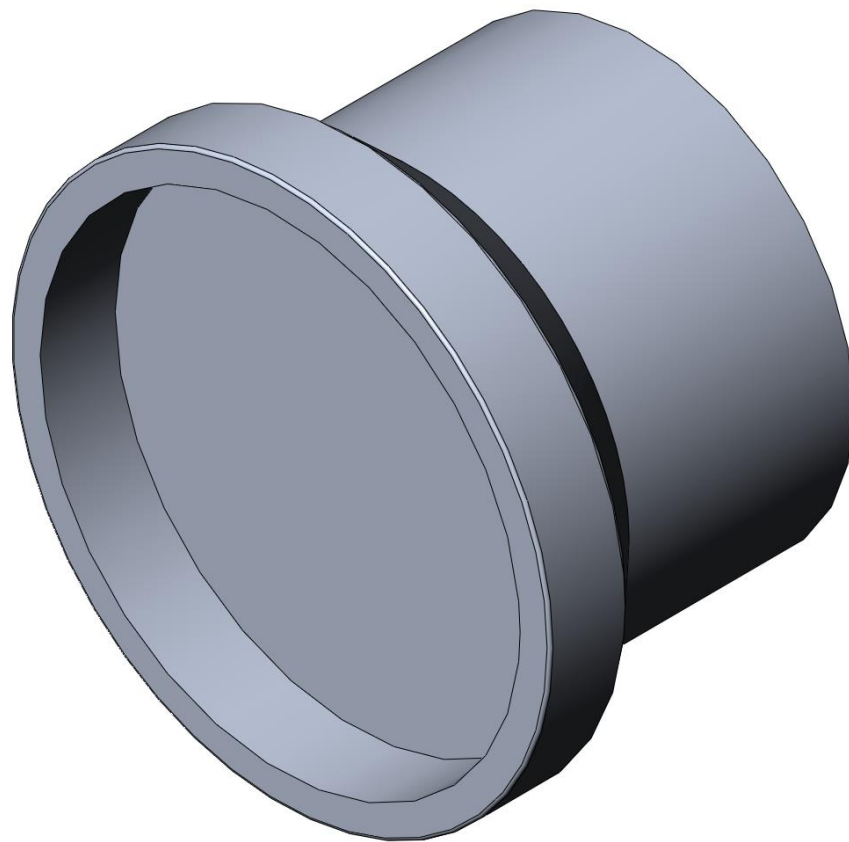


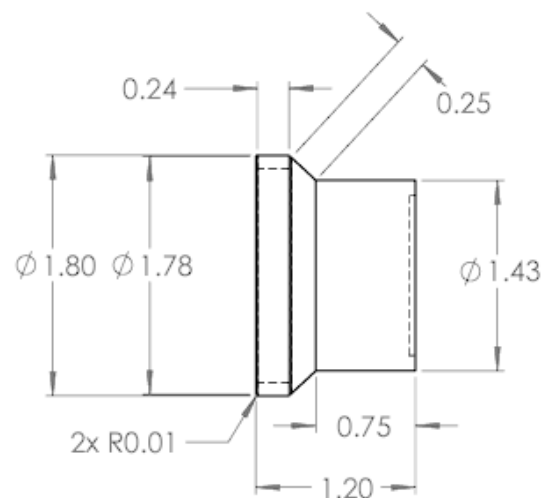
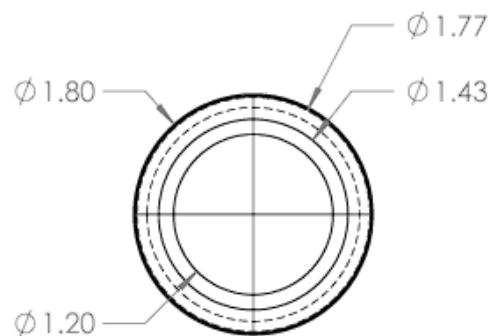
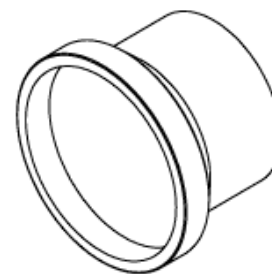
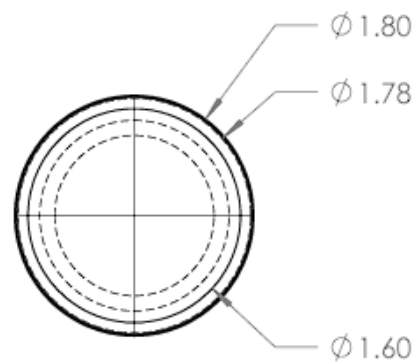


Material: Aluminum 6061
Mass: 4360.02 Kg

Scale: 1:50	Title: Telescope Lens Drawing	ENME 272 Fall 2025
Units Meter	Drawn by: L.Montilla	Date: 12/7/2025

Telescope Eyepiece (bottom)

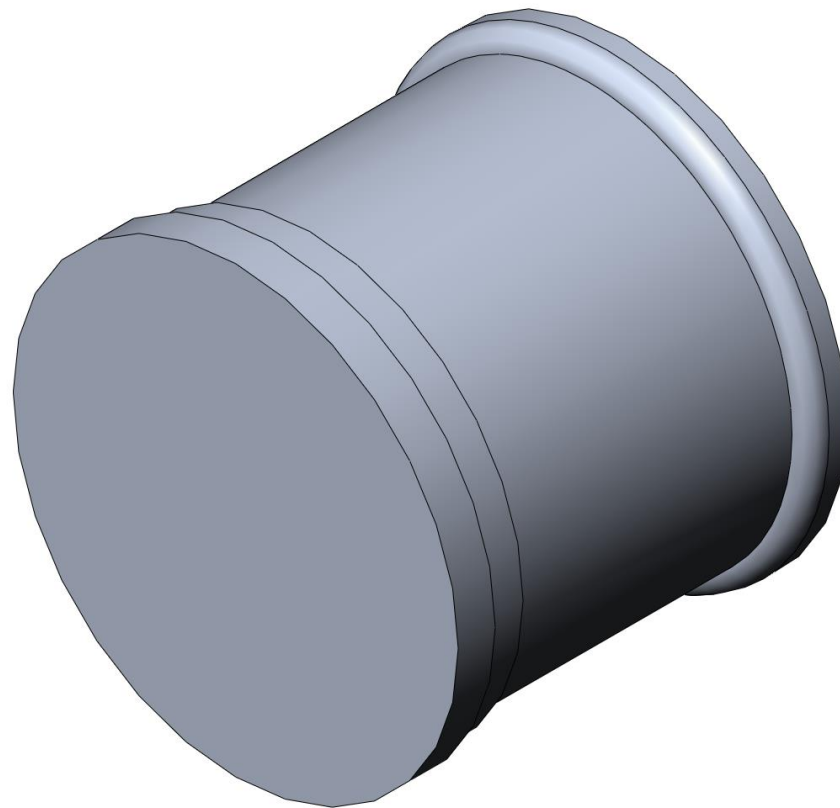


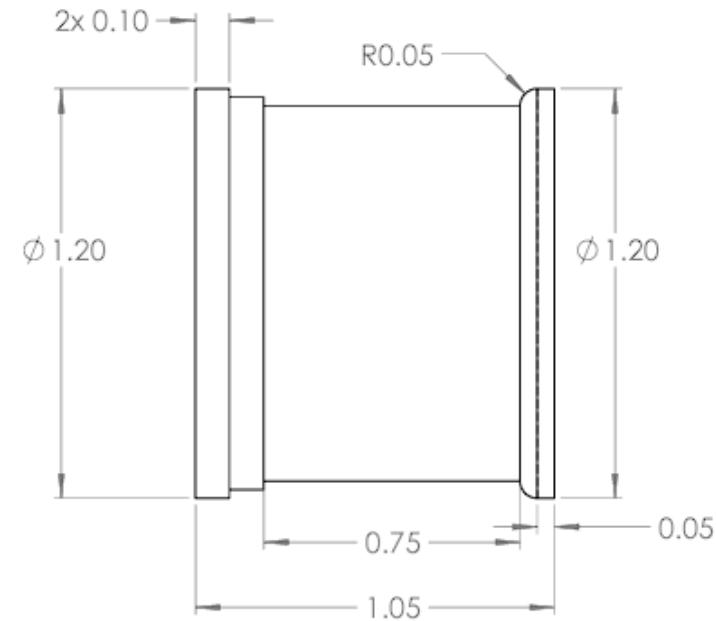
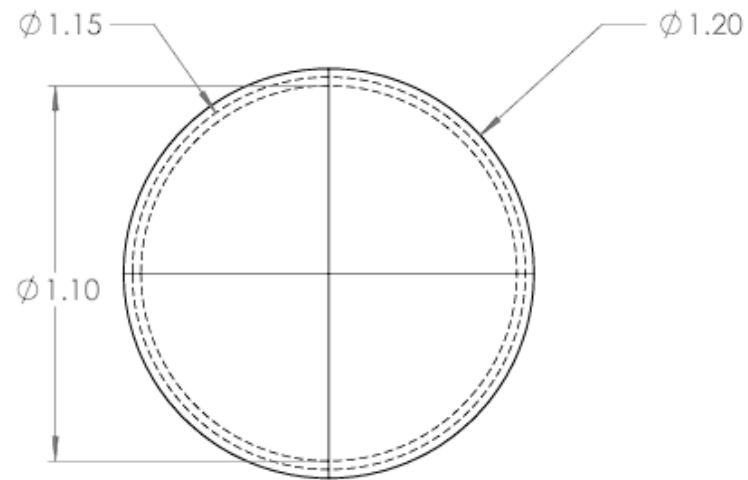
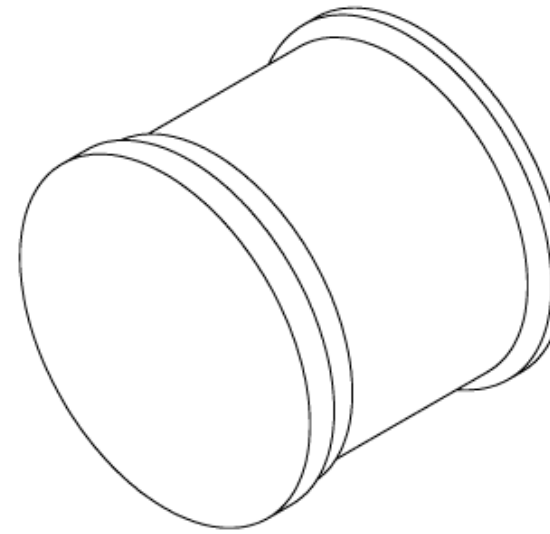
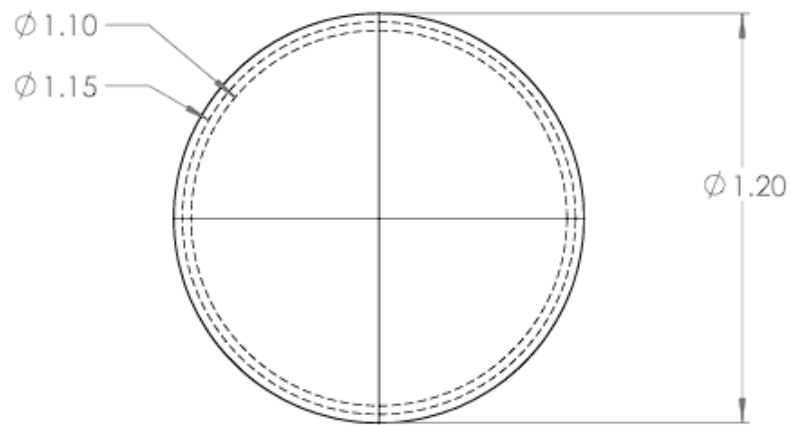


Material: Aluminum 6061
Mass: 4548.93 kg

Scale: 1:50	Title: Telescope Eyepiece bottom	ENME 272 Fall 2025
Units Meter	Drawn by: L.Montilla	Date: 12/7/2025

Telescope Eyepiece (Top)

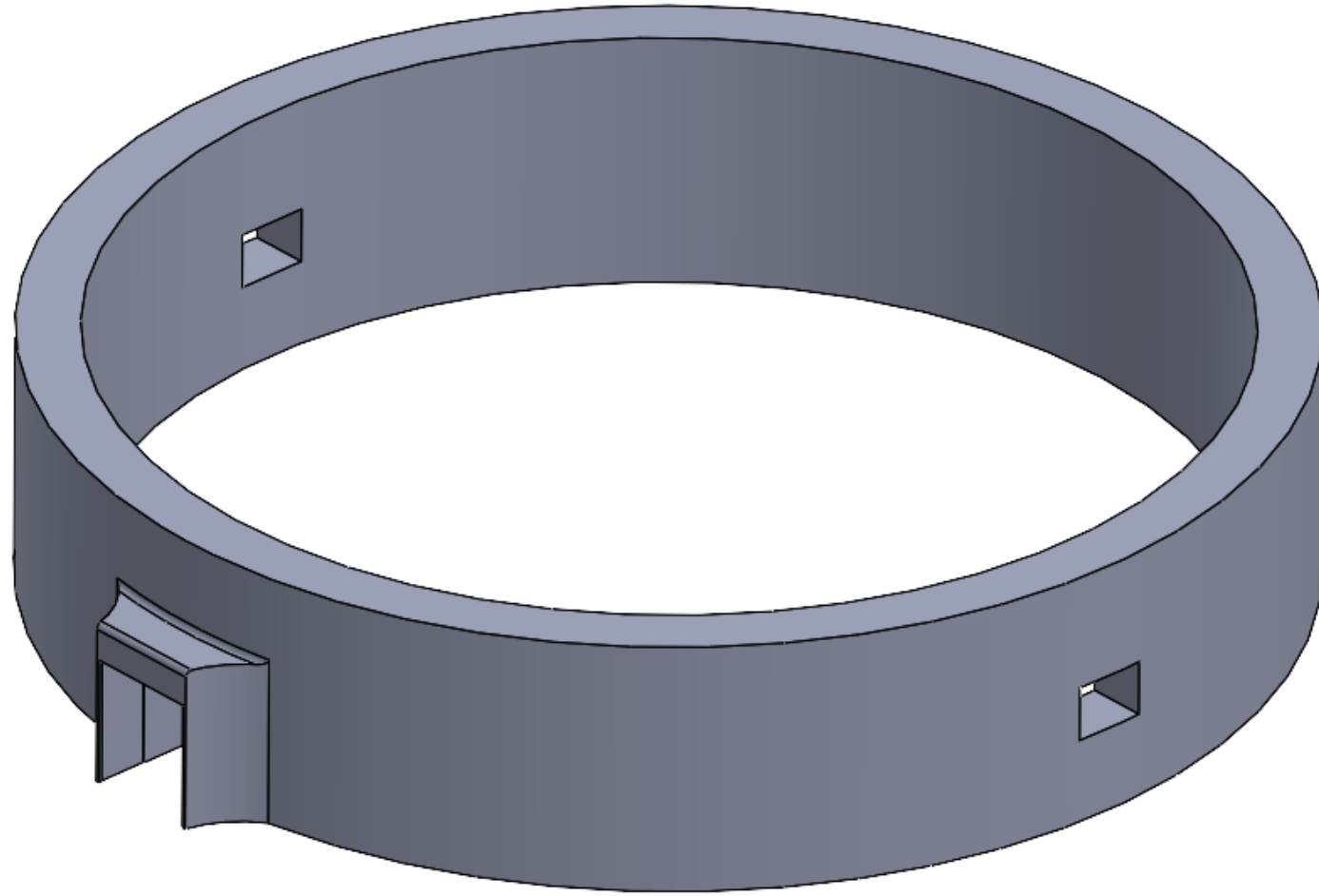


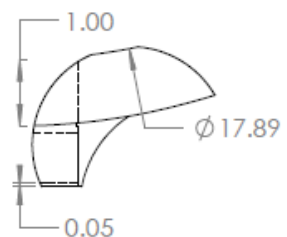
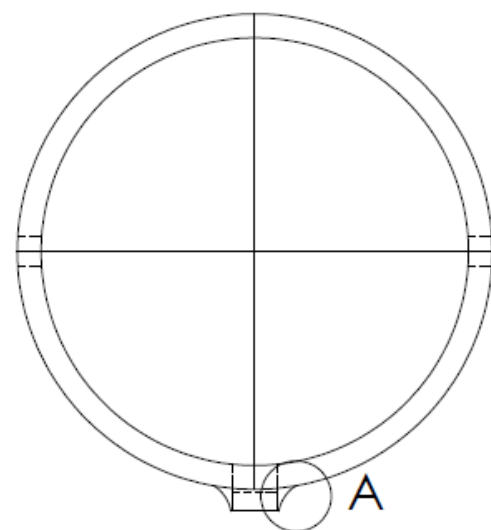


Material: Aluminum 6061
Mass: 2810.23 Kg

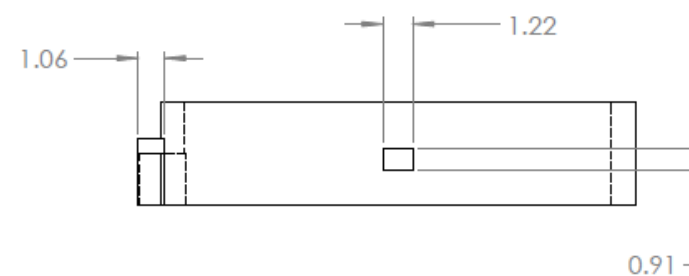
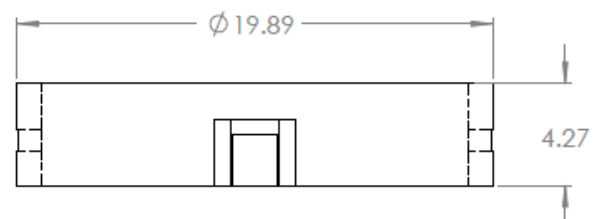
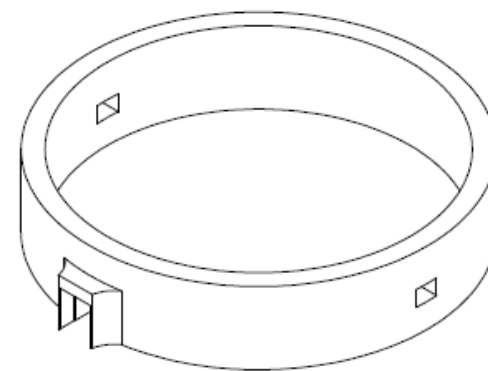
Scale: 1:20	Title: Eyepiece top	ENME 272 Fall 2025
Units Meter	Drawn by: L.Montilla	Date: 12/7/2025

Observatory Base



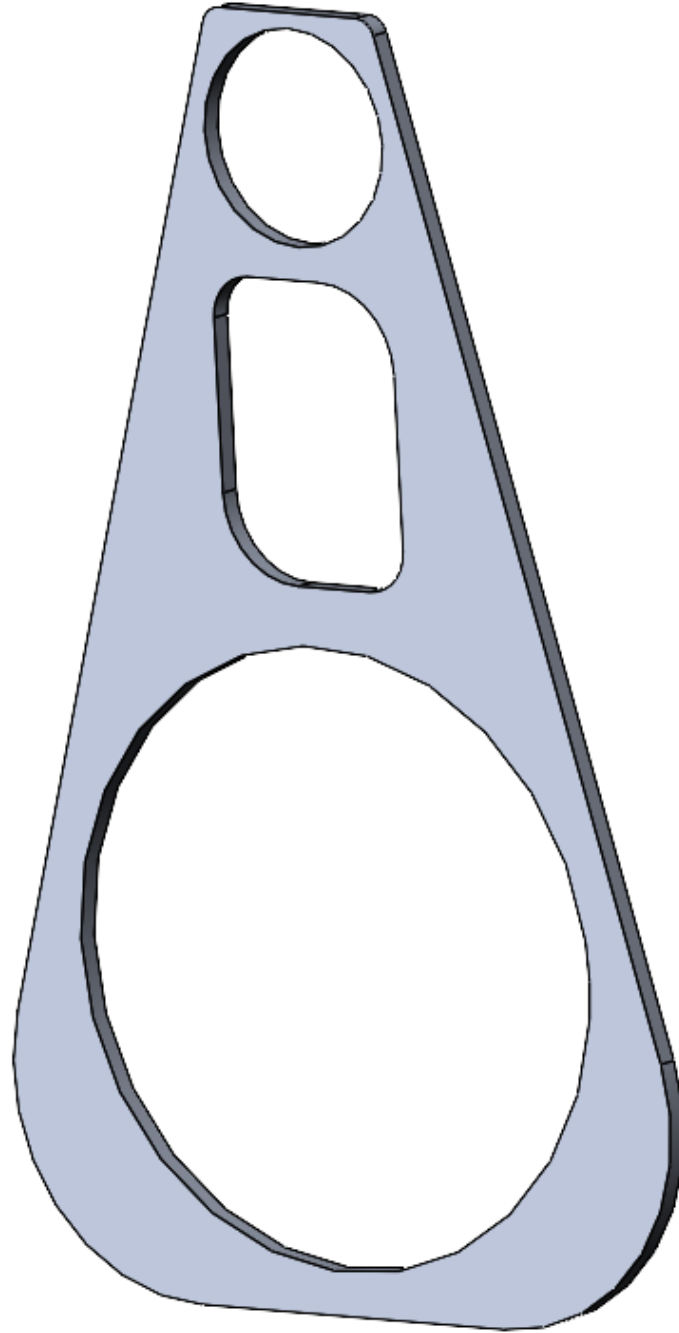


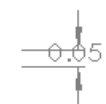
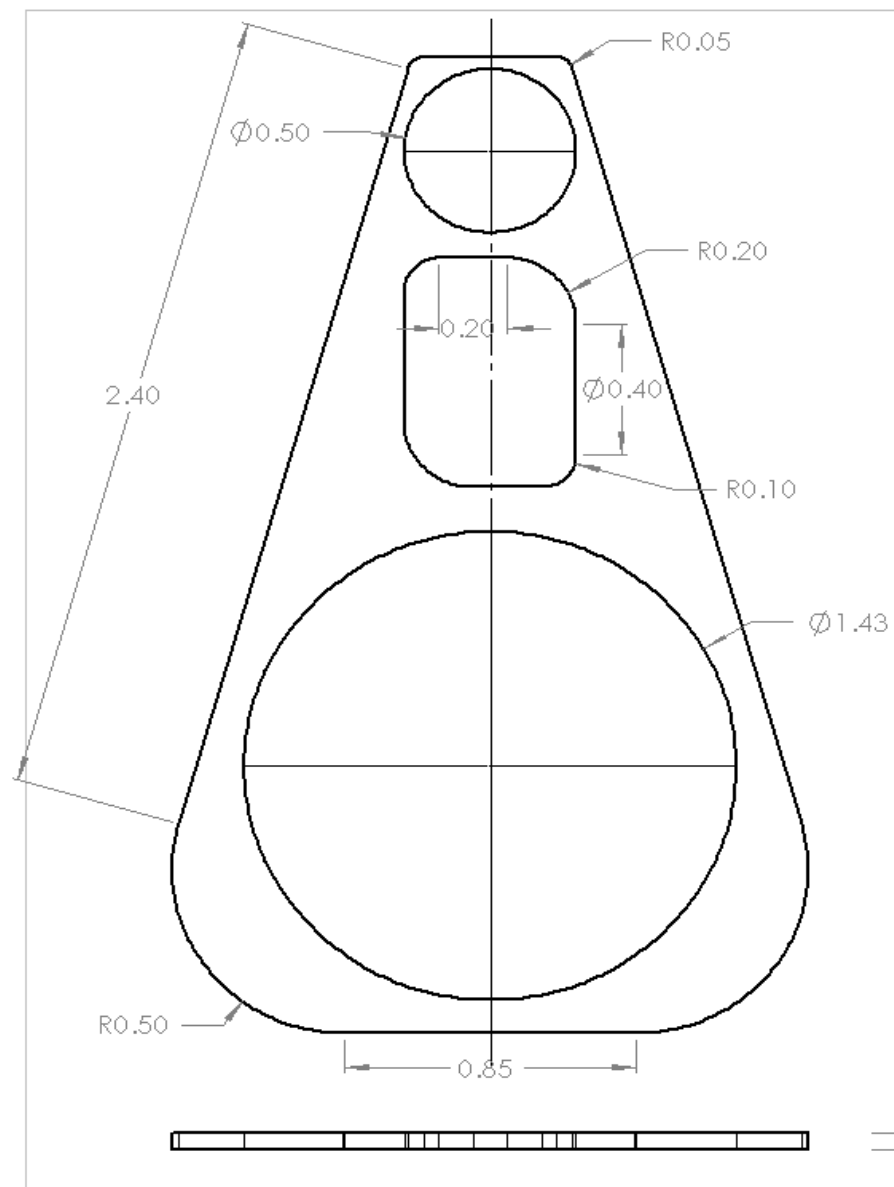
DETAIL A
SCALE 1 : 90



Scale: 1:250	Title: observatory base drawing	ENME 272 Fall 2025
Units meters	Drawn by: Isabella Scianablo	Date: 12/7/2025

Connector

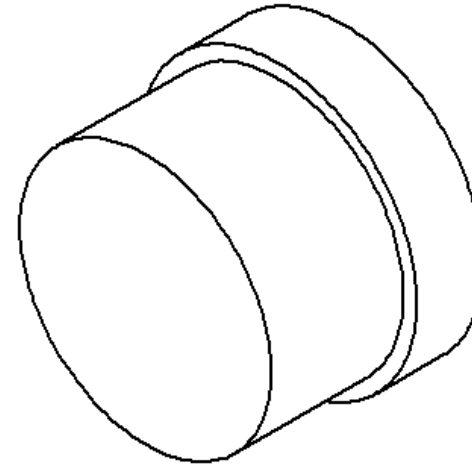
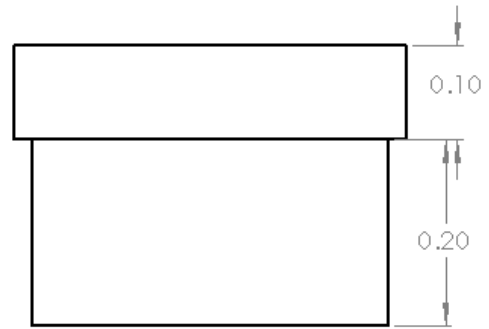




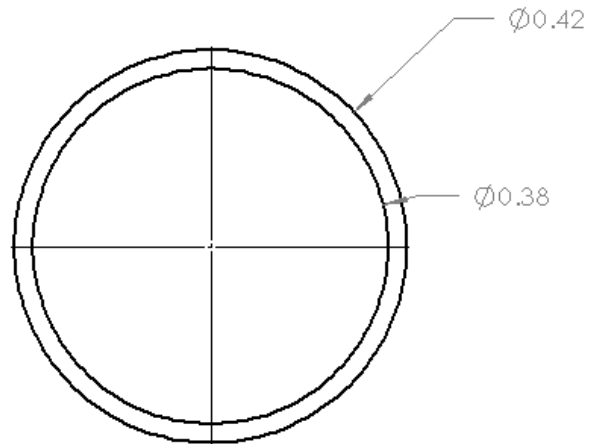
Scale: 1:20	Title: connector	ENME 272 Fall 2025
Units mks	Drawn by: amanda lau	Date: 12/8/2025

End Piece



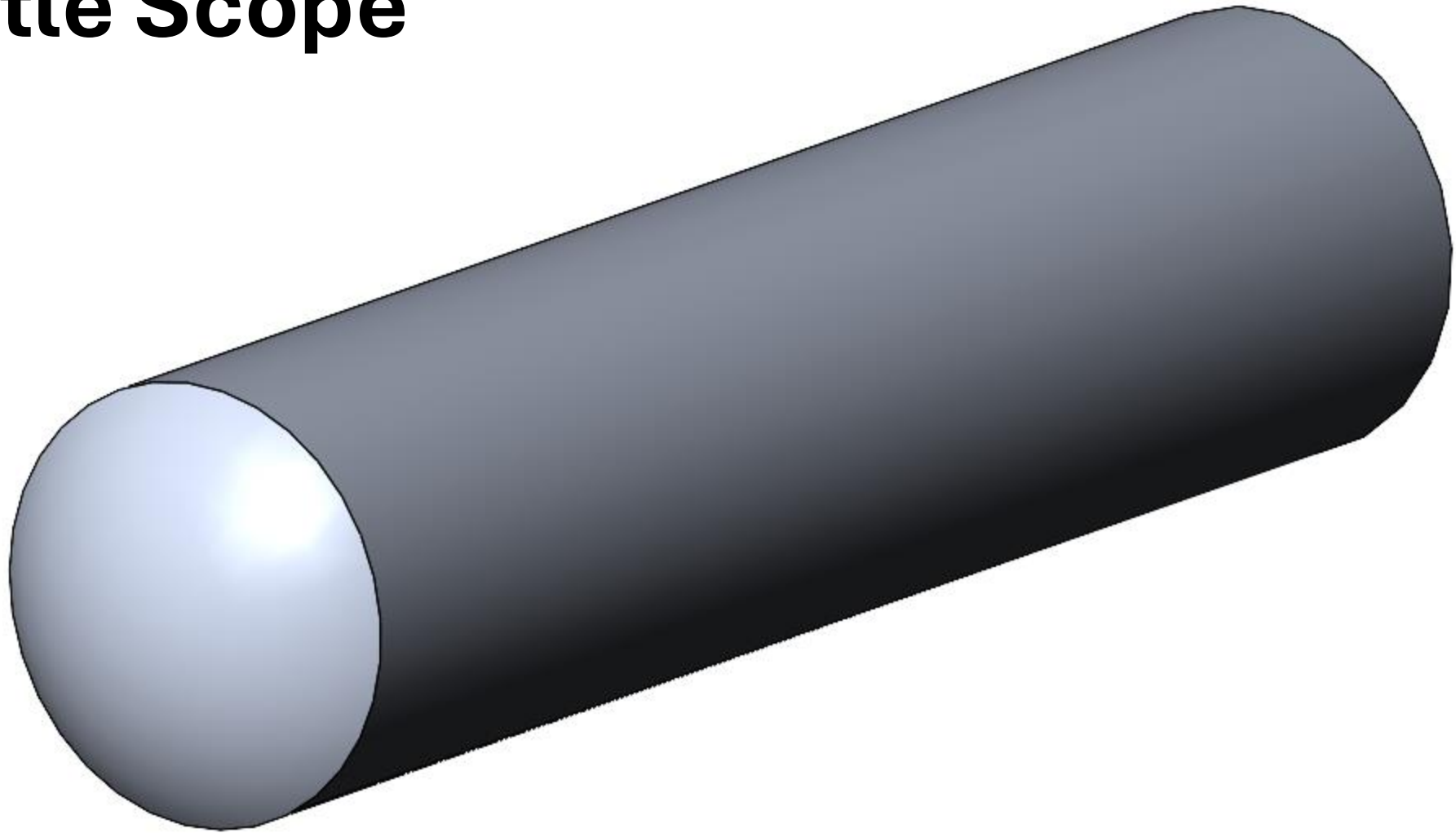


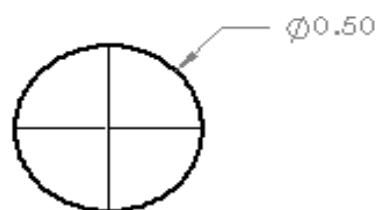
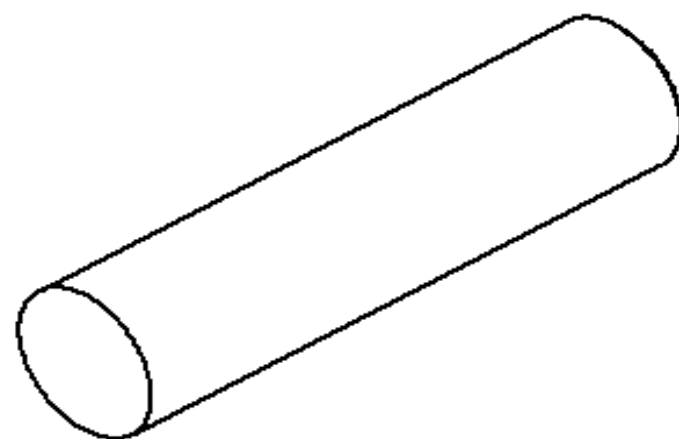
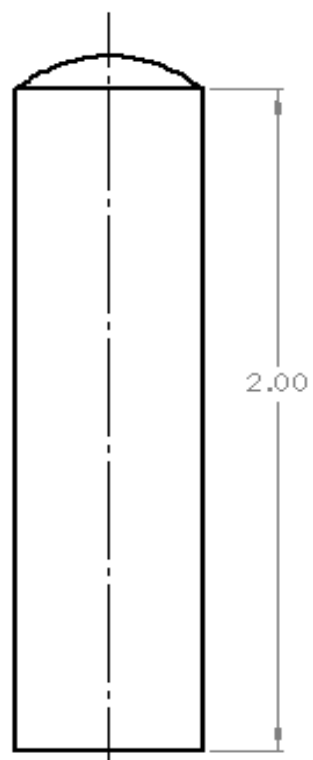
material: aluminum 6061



Scale: 1:7	Title: end	ENME 272 Fall 2025
Units mks	Drawn by: amanda lau	Date: 12/8/2025

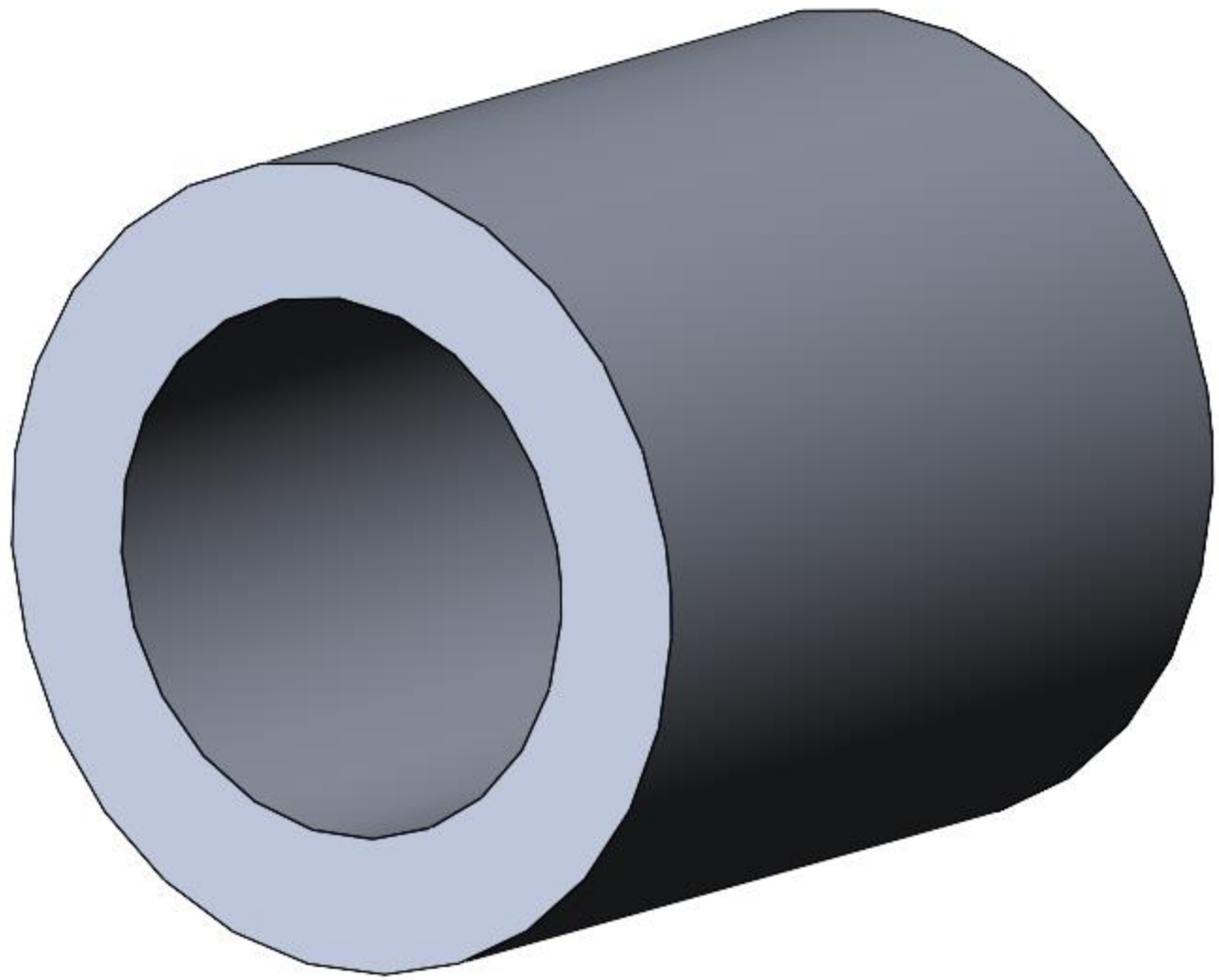
Little Scope

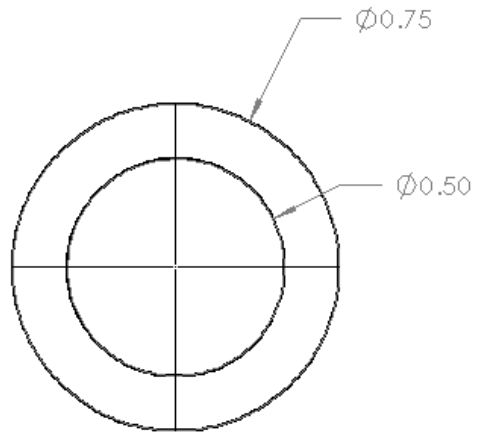
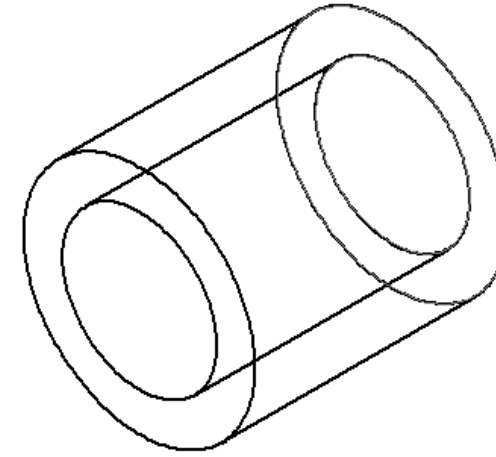
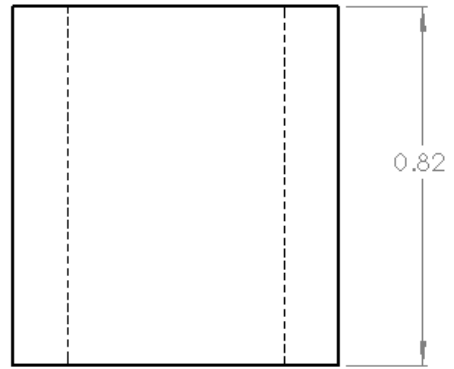




Scale: 1:20	Title: scope little	EN ME 272 Fall 2025
Units inch	Drawn by: amanda lau	Date: 12/8/2025

Middle Ring



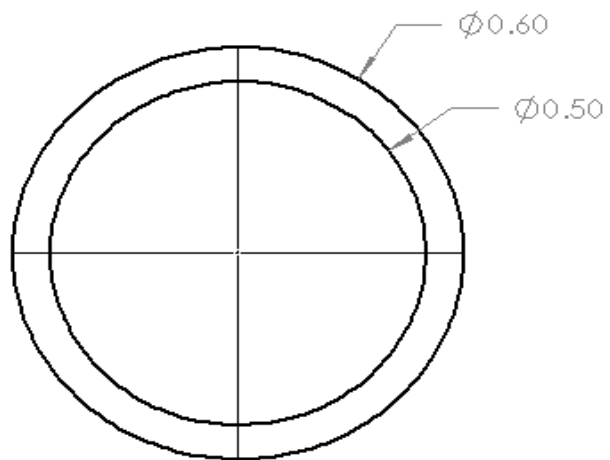
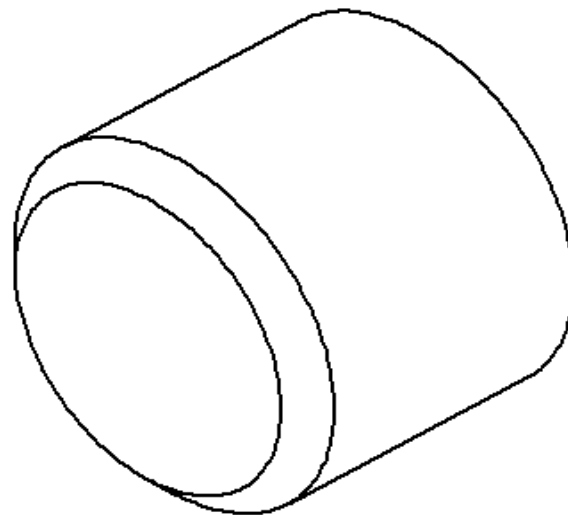
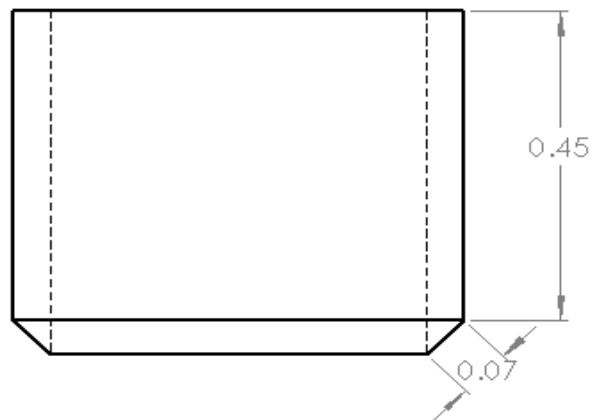


material: aluminum 6061

Scale: 1:15	Title: middle ring	ENME 272 Fall 2025
Units mks	Drawn by: amanda lau	Date: 12/8/2025

Lens

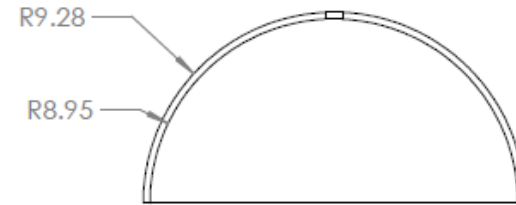
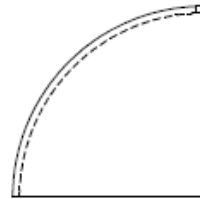
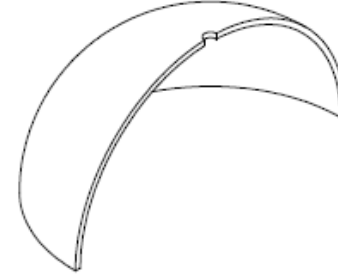
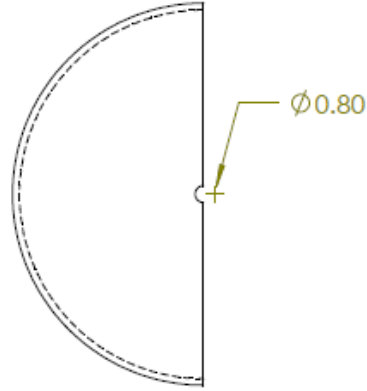




Scale: 1:10	Title: lens	ENME 272 Fall 2025
Units mks	Drawn by: amanda lau	Date: 12/8/2025

Inner Dome

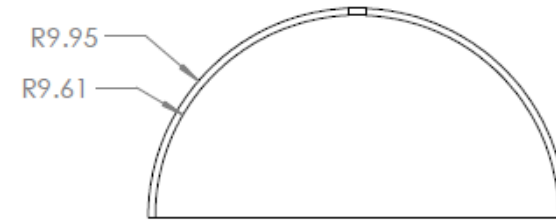
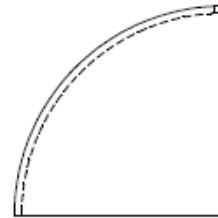
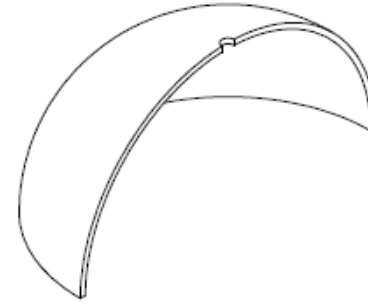
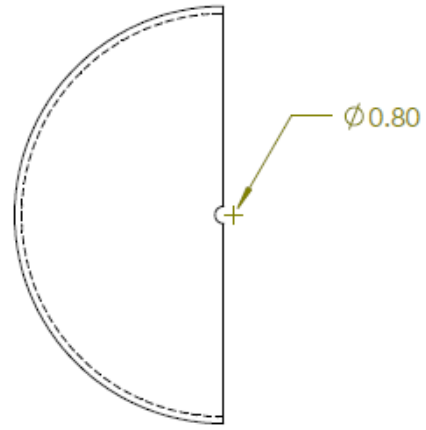




Scale: 1:300	Title: inner dome	ENME 272 Fall 2025
Units meters	Drawn by: Isabella Scianablo	Date: 12/8/2025

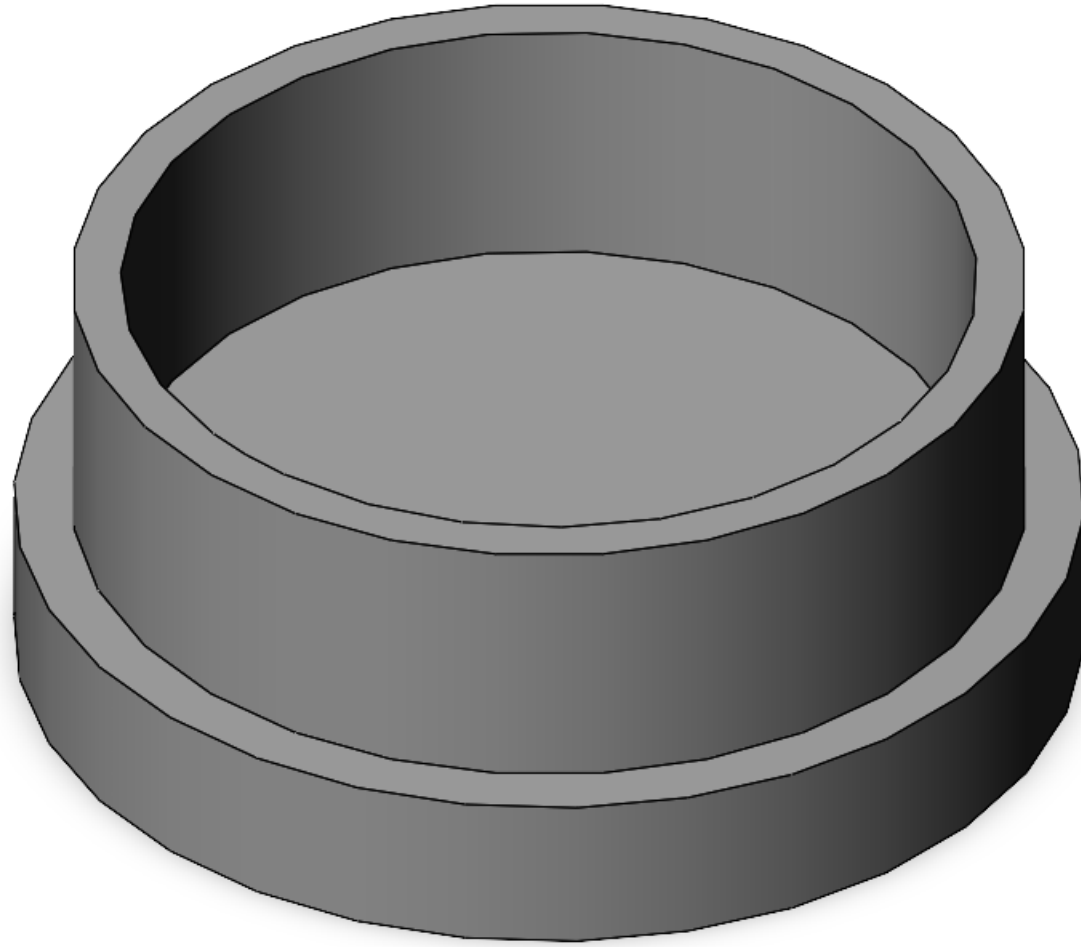
Outer Dome

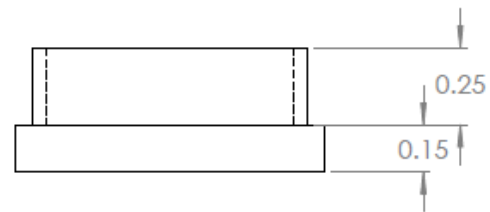
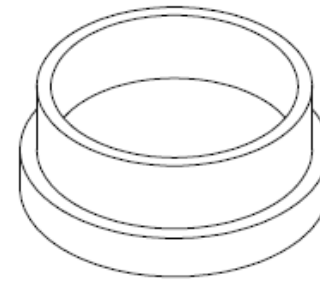
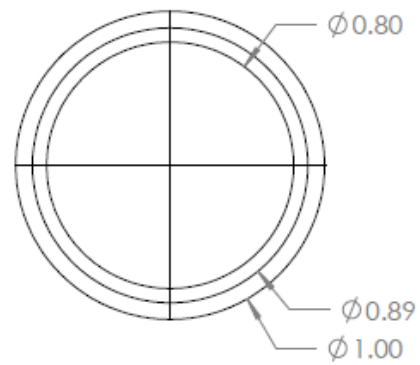




Scale: 1:300	Title: outer dome drawing	ENME 272 Fall 2025
Units meters	Drawn by: Isabella Scianablo	Date: 12/8/2025

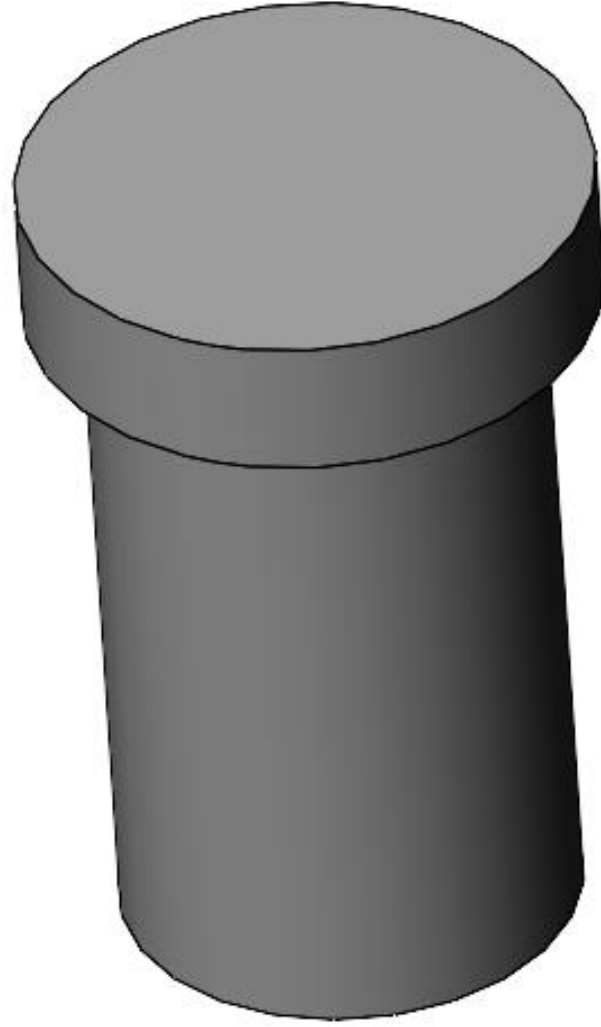
Pin Cap

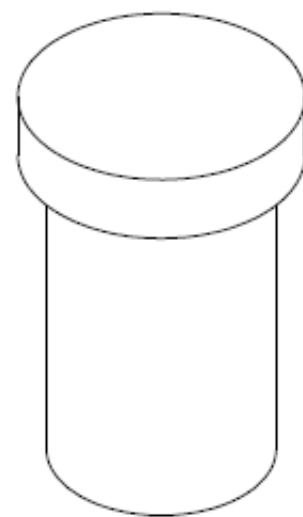
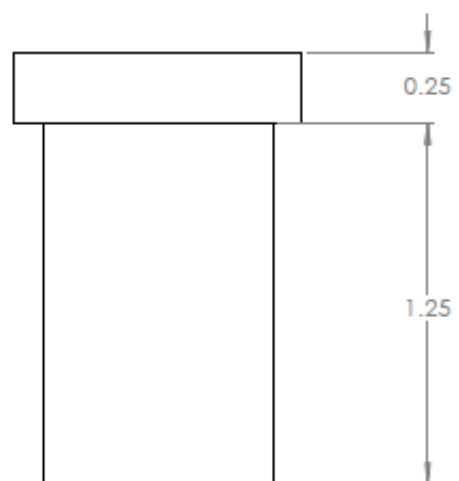
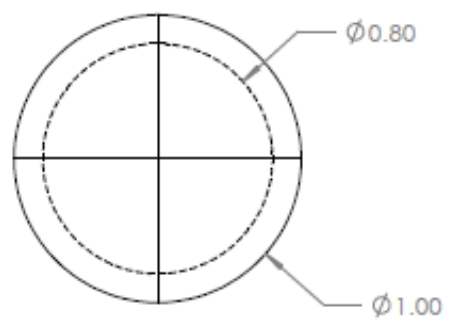




Scale: 1:20	Title: pin cap drawing	ENME 272 Fall 2025
Units meters	Drawn by: Isabella Scianablo	Date: 12/7/2025

Pin

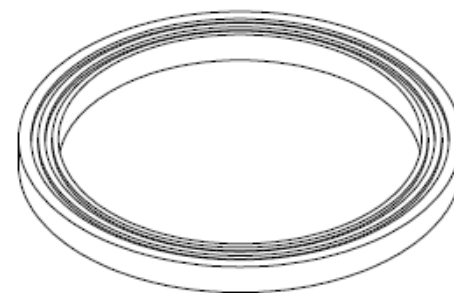
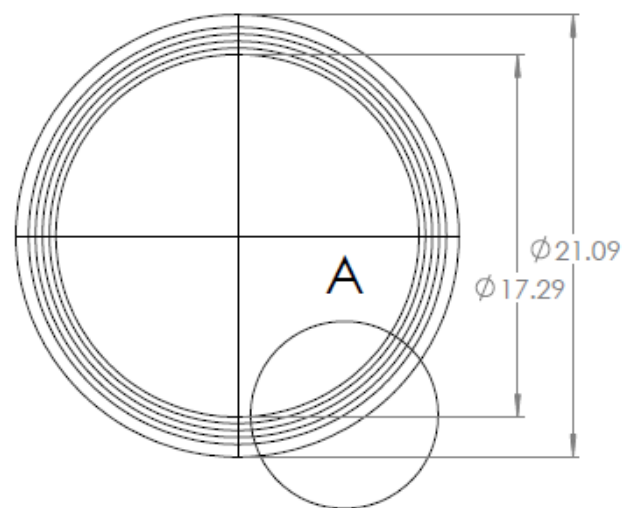




Scale: 1:20	Title: pin drawing	ENME 272 Fall 2025
Units meters	Drawn by: Isabella Scianablo	Date: 12/8/2025

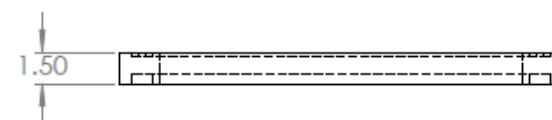
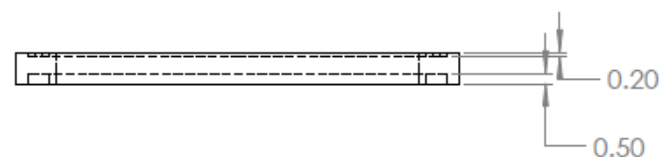
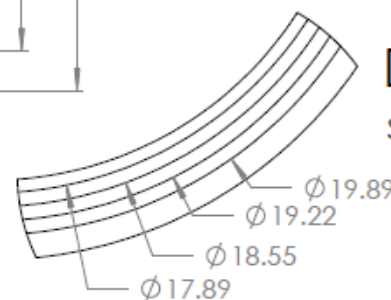
Dome Holder





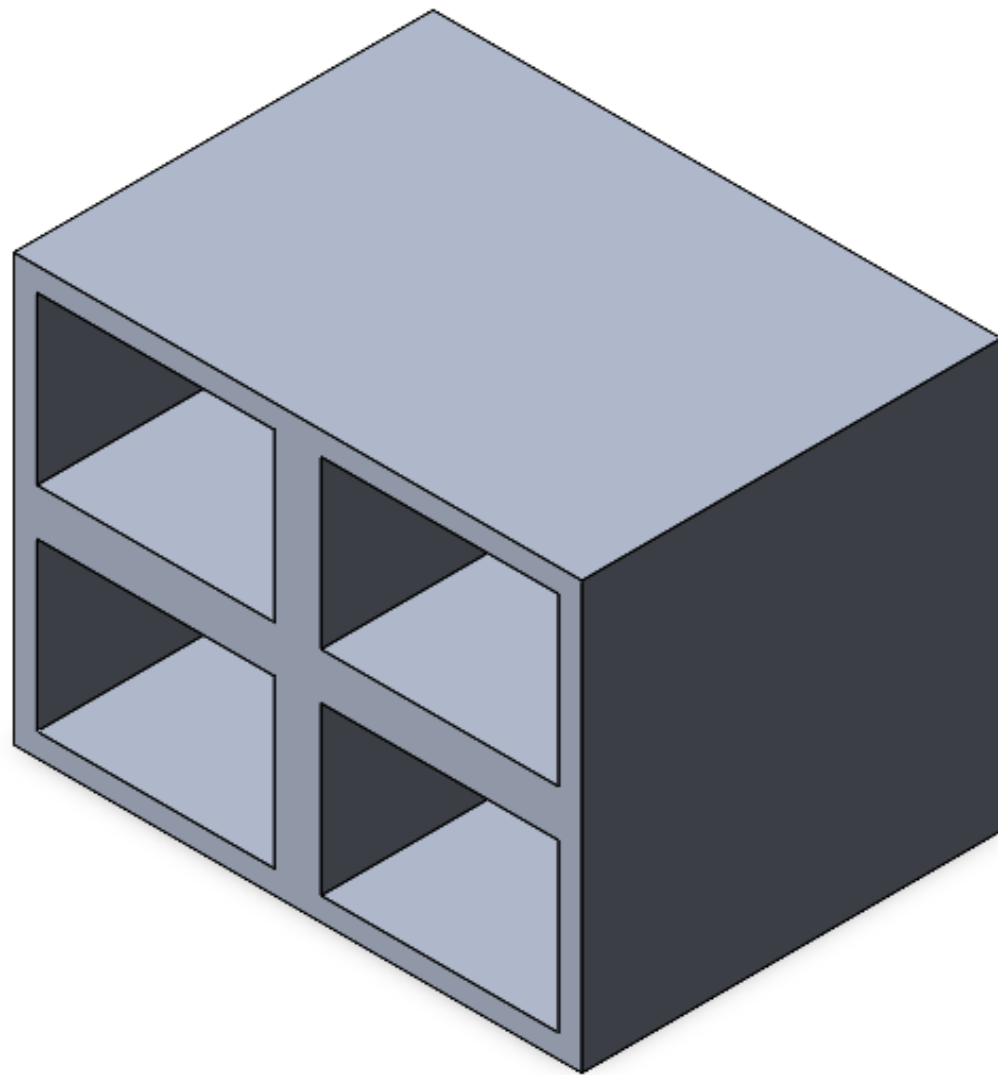
DETAIL A

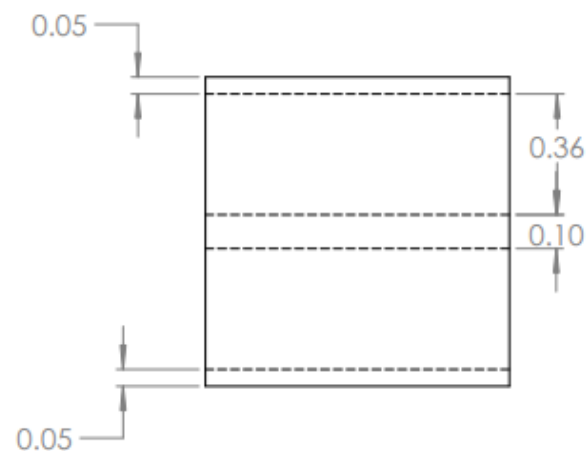
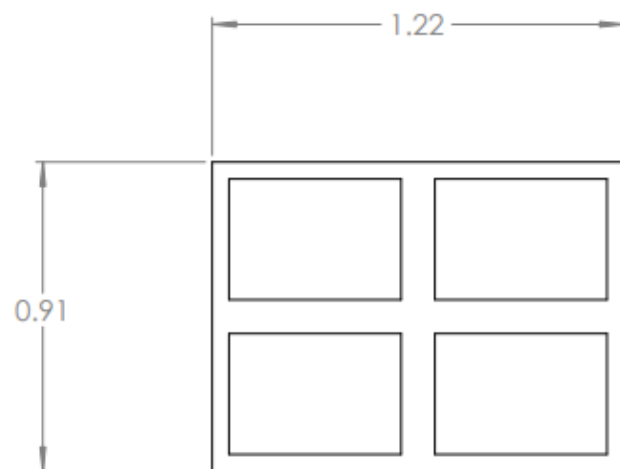
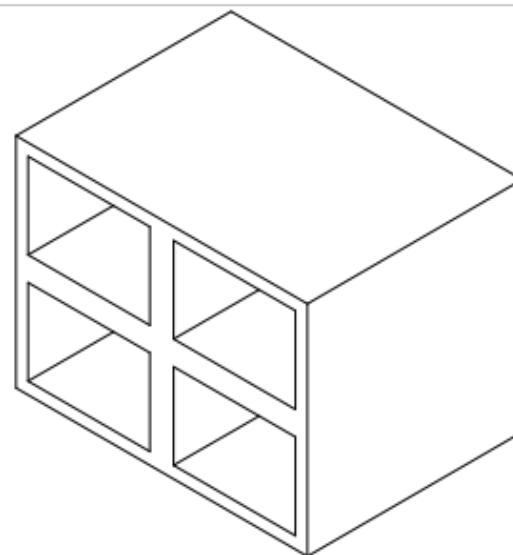
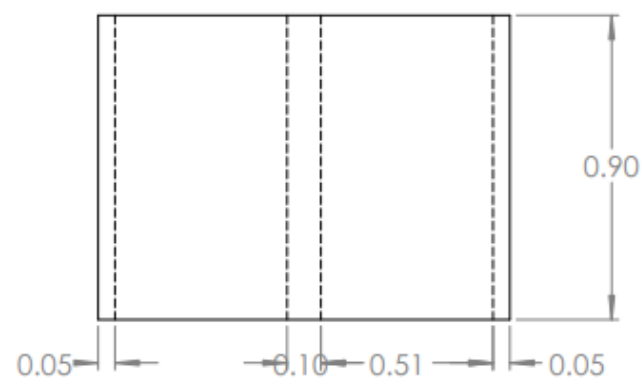
SCALE 1 : 150



Scale: 1:300	Title: dome holder final drawing	ENME 272 Fall 2025
Units meters	Drawn by: Isabella Scianablo	Date: 12/8/2025

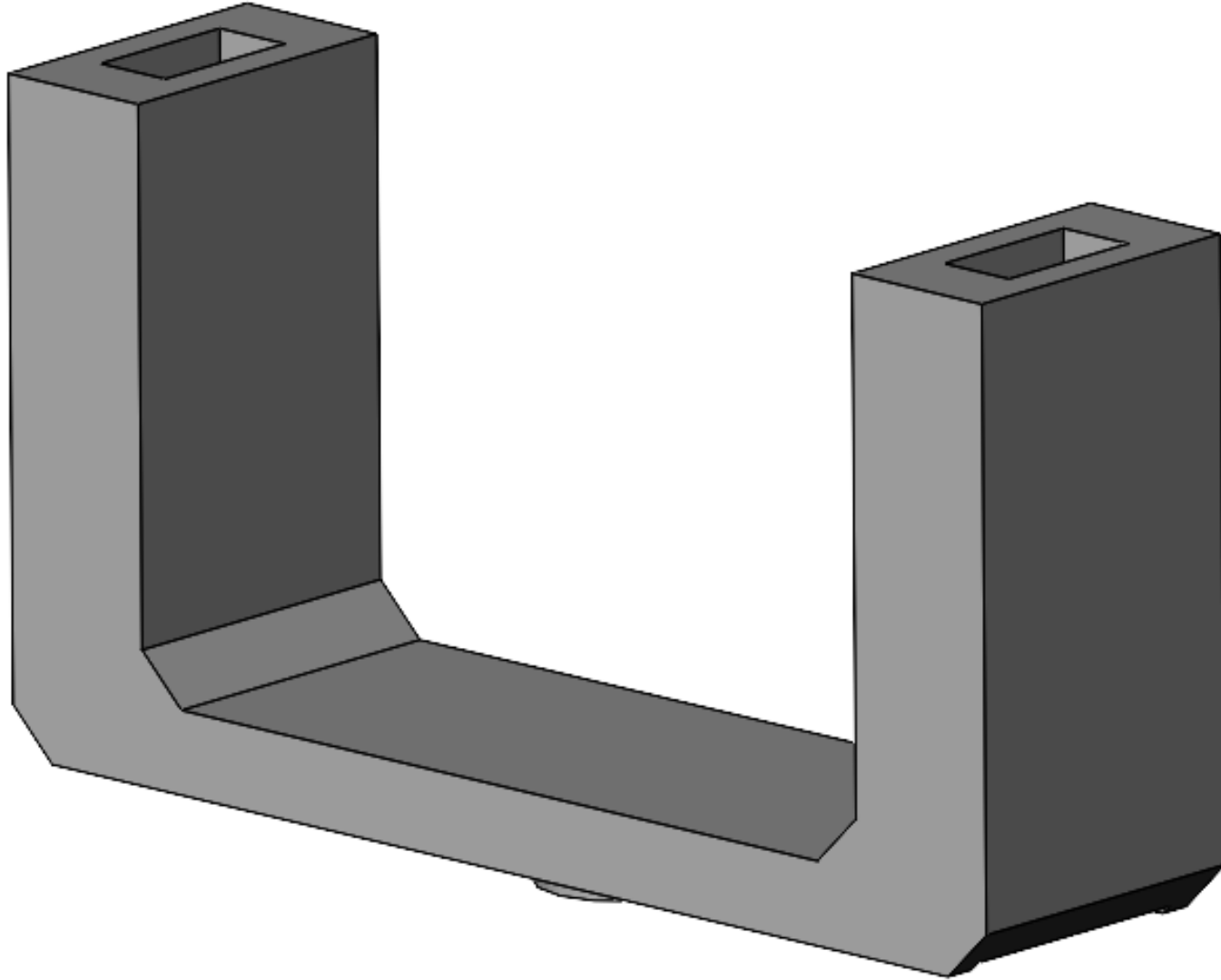
Window

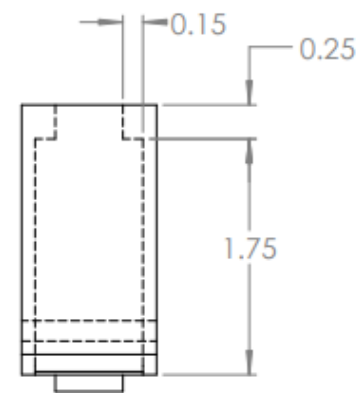
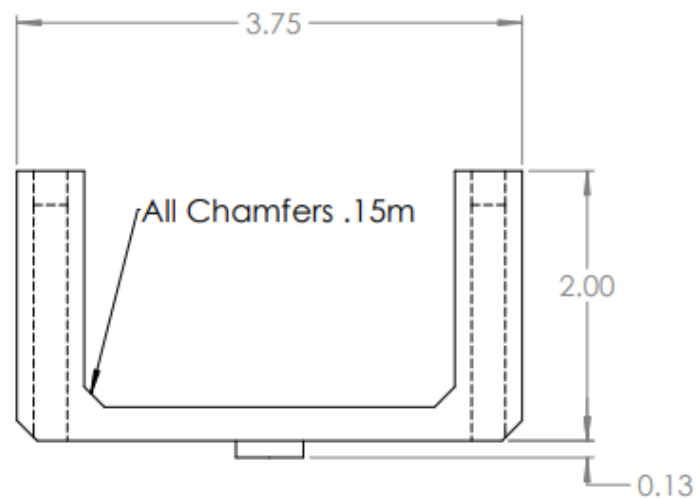
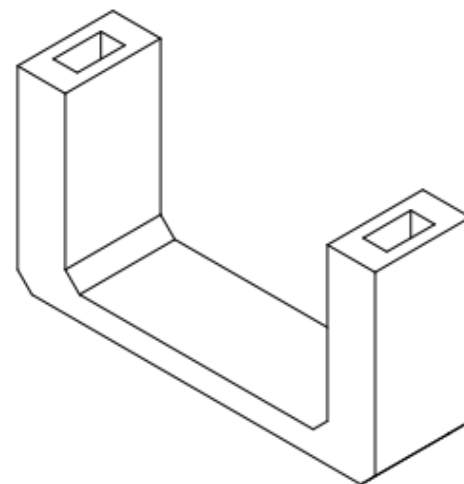
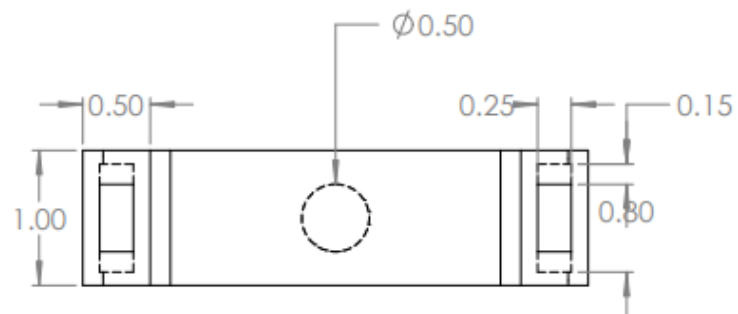




Scale: 1:20	Title: Window	ENME 272 Fall 2025
Units Meters	Drawn by: Hiro Okamoto	Date: 12/8/2025

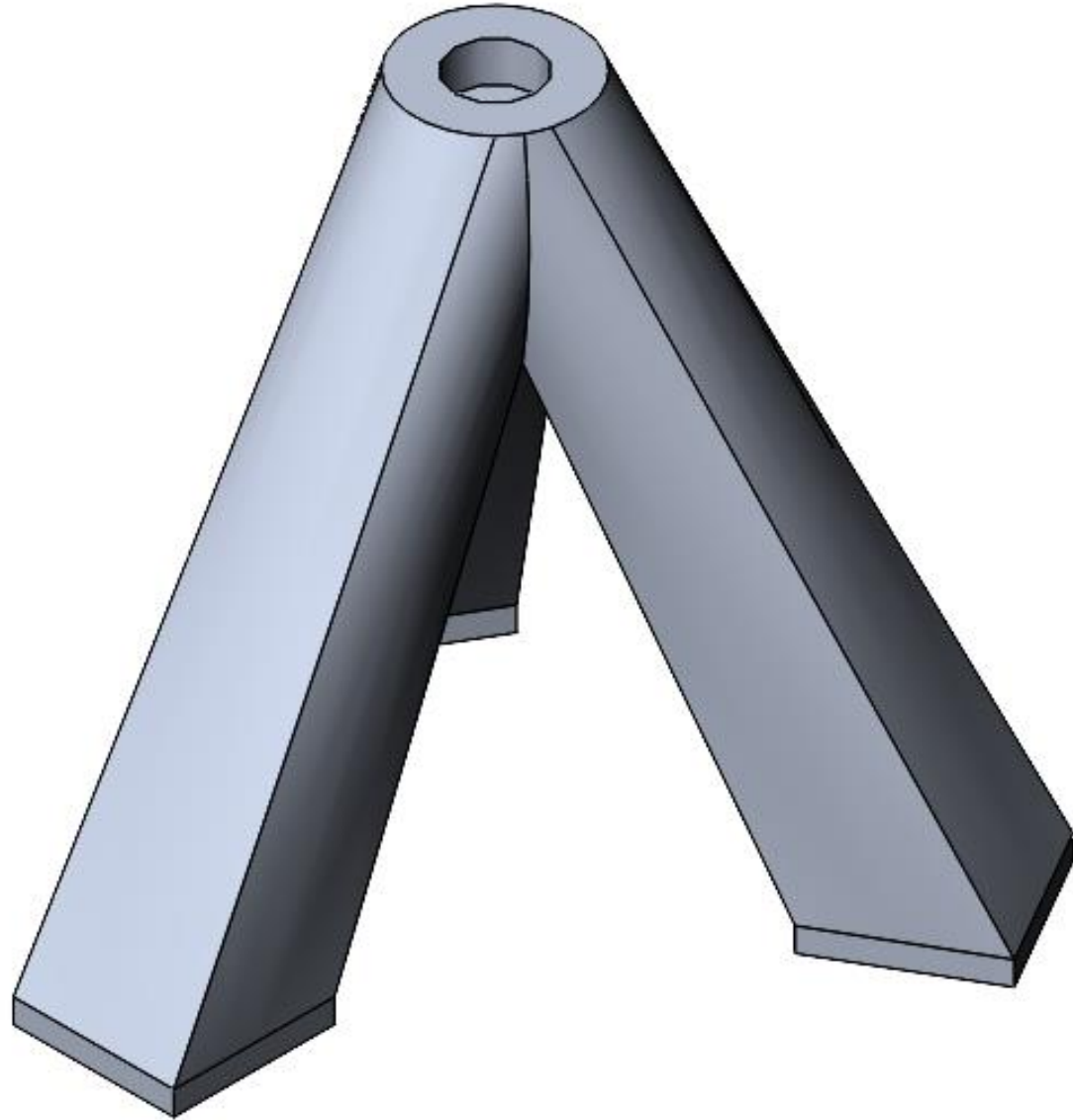
Telescope Holder

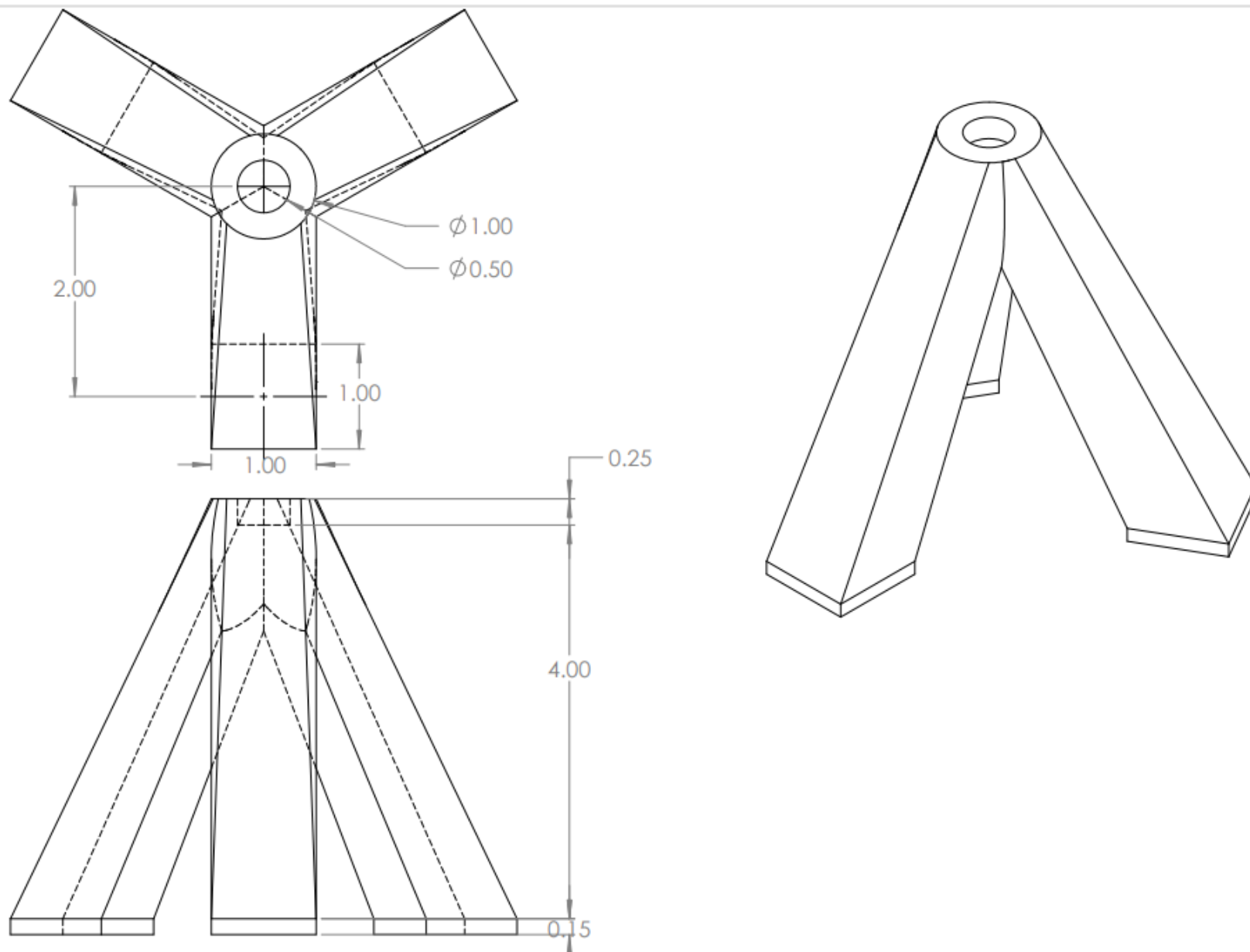




Scale: 1:50	Title: Telescope holder	ENME 272 Fall 2025
Units Meters	Drawn by: Hiro Okamoto	Date: 12/8/2025

Stand





Scale:
1:50

Title:
Stand

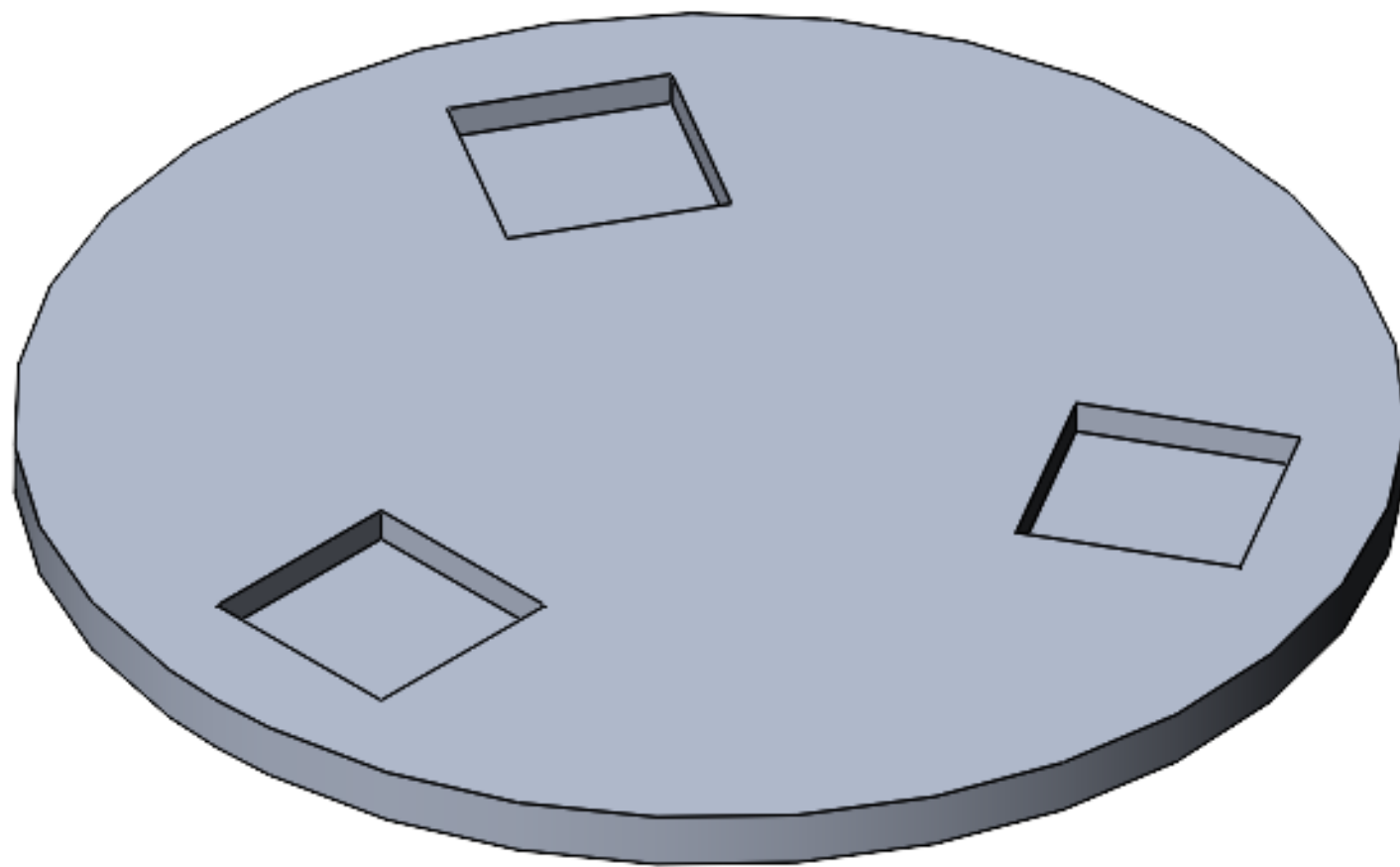
ENME 272
Fall 2025

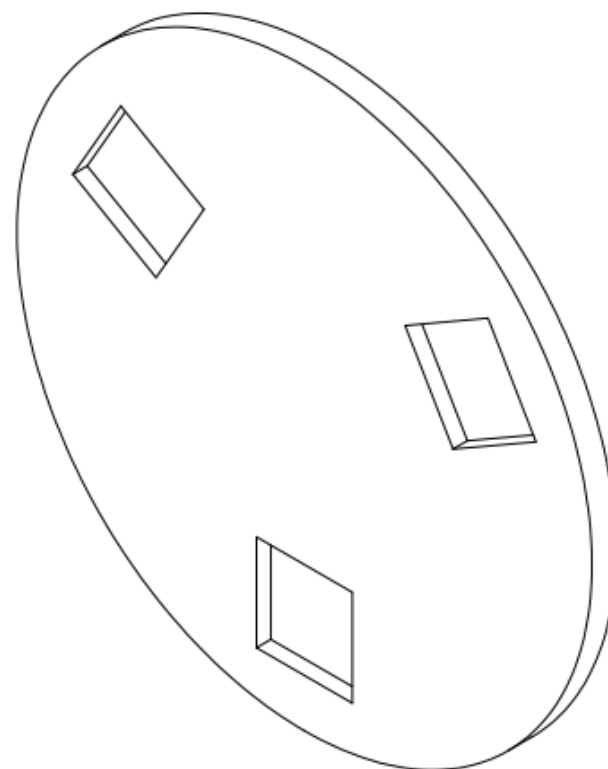
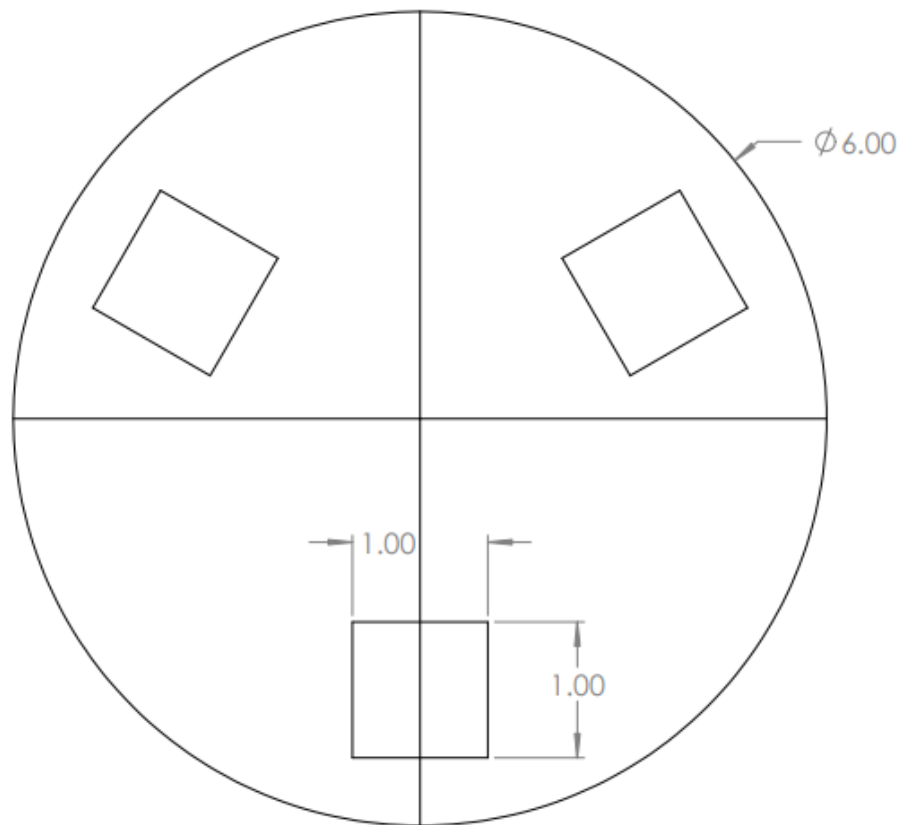
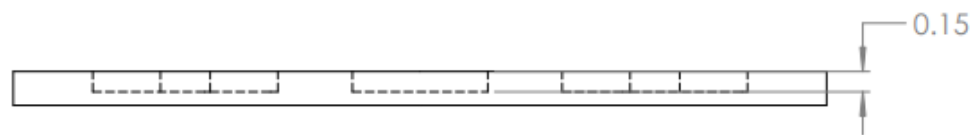
Units
Meters

Drawn by:
Hiro Okamoto

Date:
12/7/2025

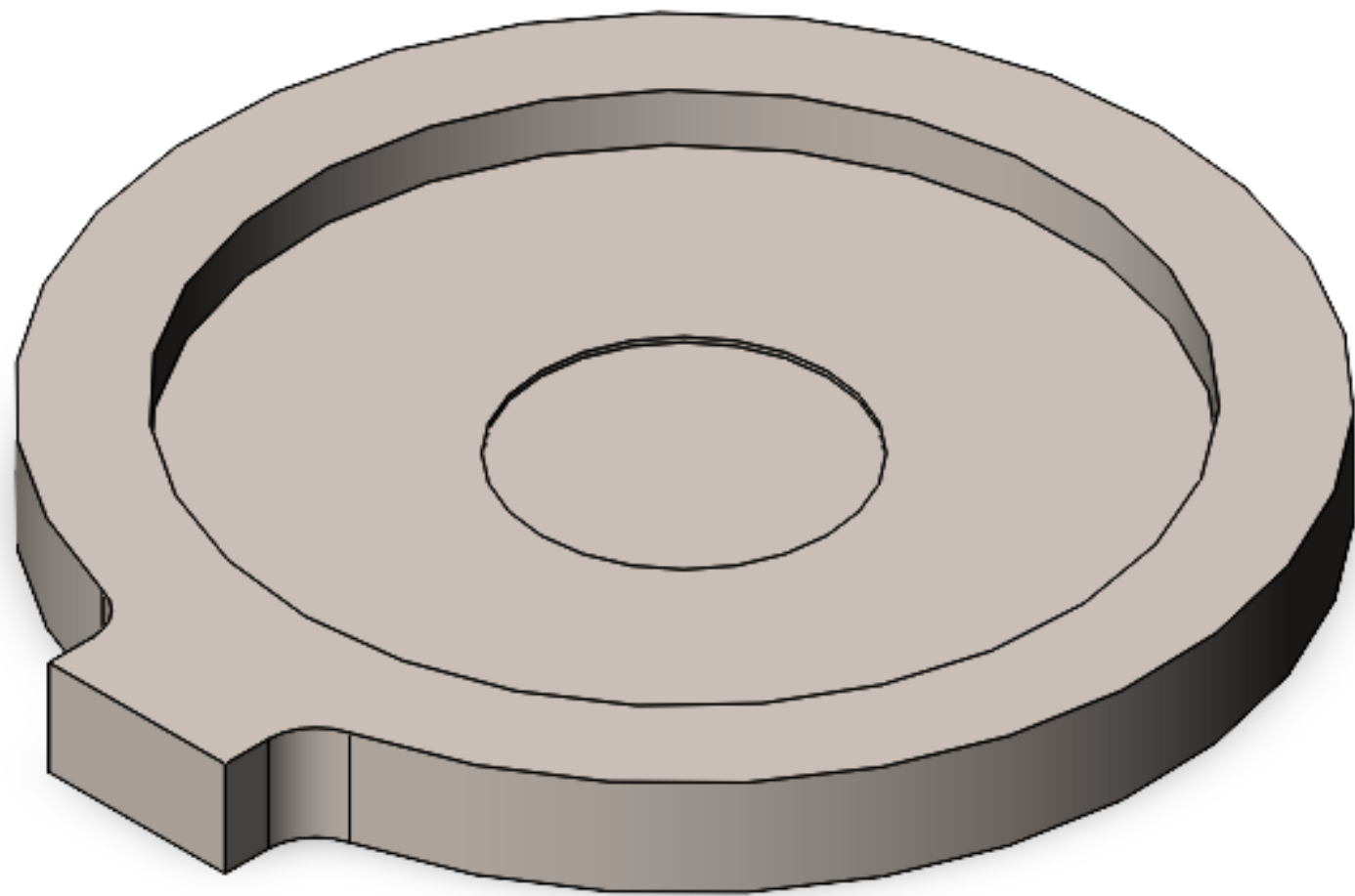
Base for Stand

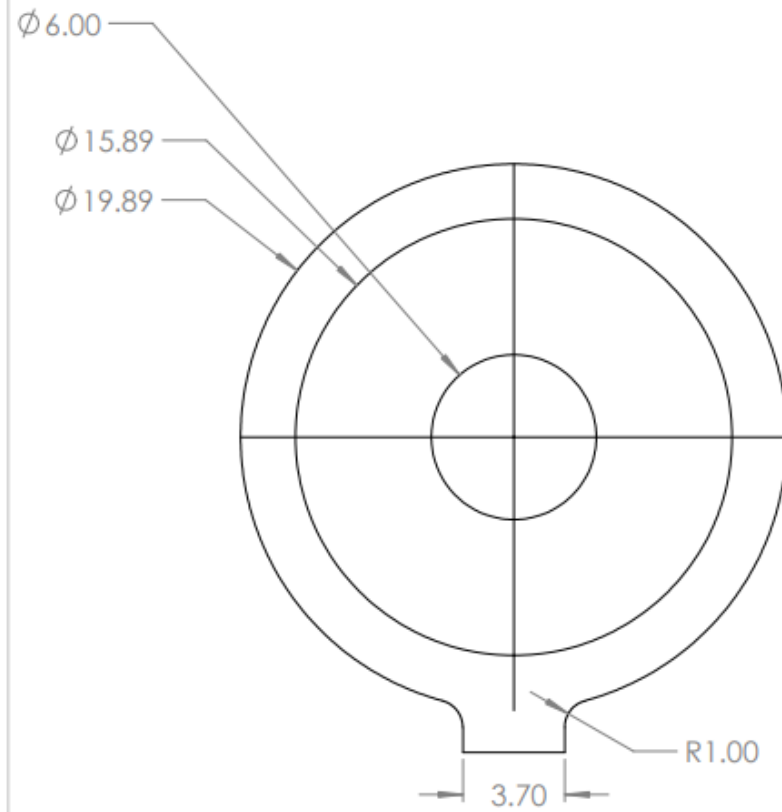
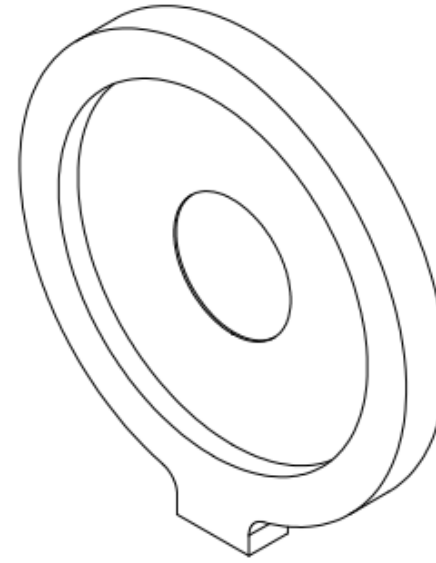
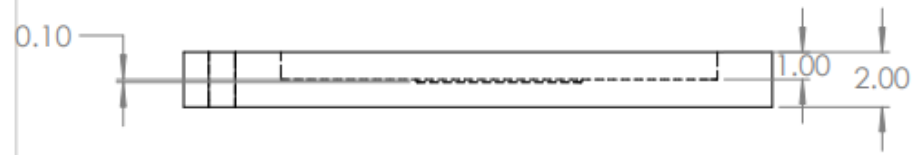




Scale: 1:50	Title: Base	ENME 272 Fall 2025
Units Meters	Drawn by: Hiro Okamoto	Date: 12/7/2025

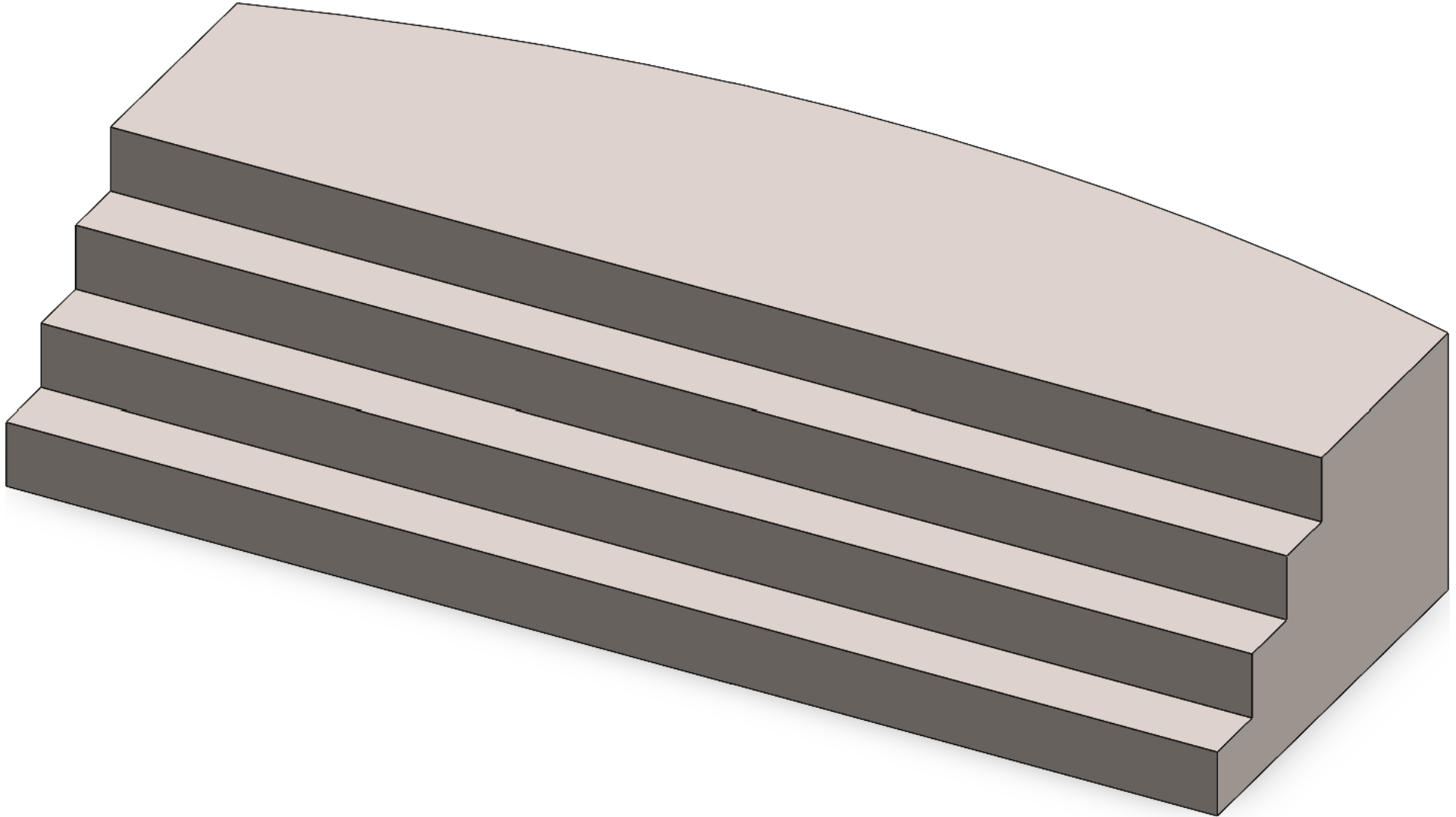
Floor

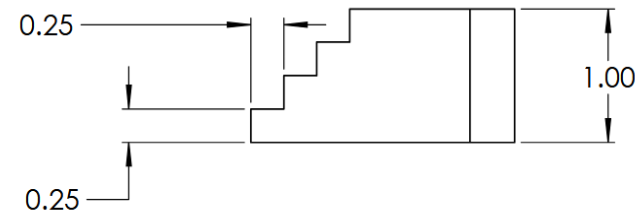
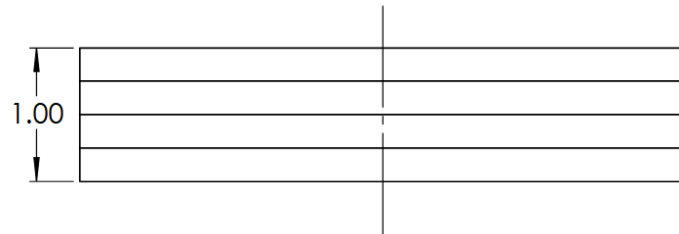
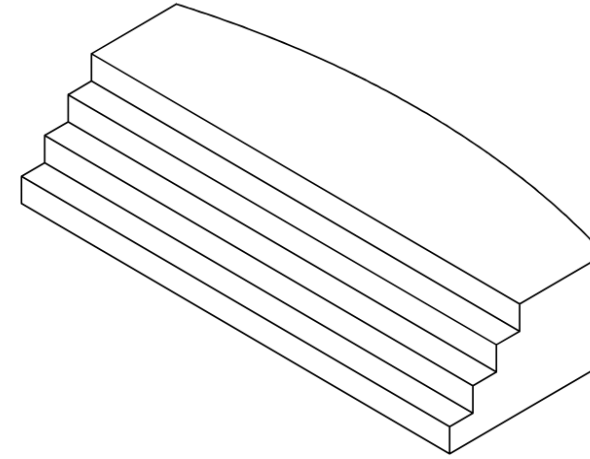
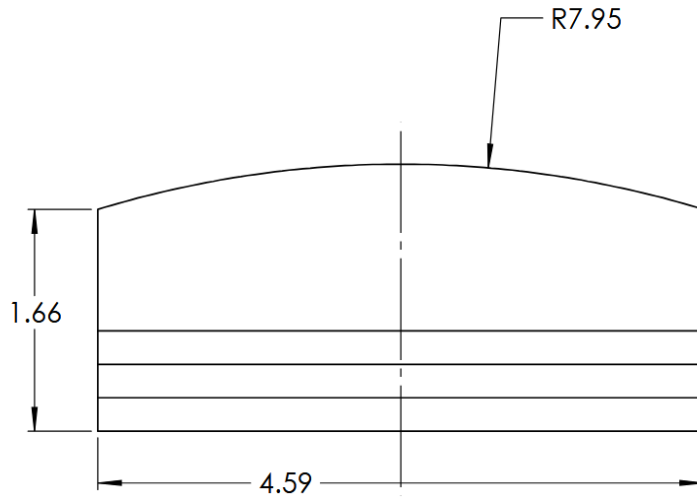




Scale: 1:250	Title: Floor	ENME 272 Fall 2025
Units Meters	Drawn by: Hiro Okamoto	Date: 12/7/2025

Inside Stairs



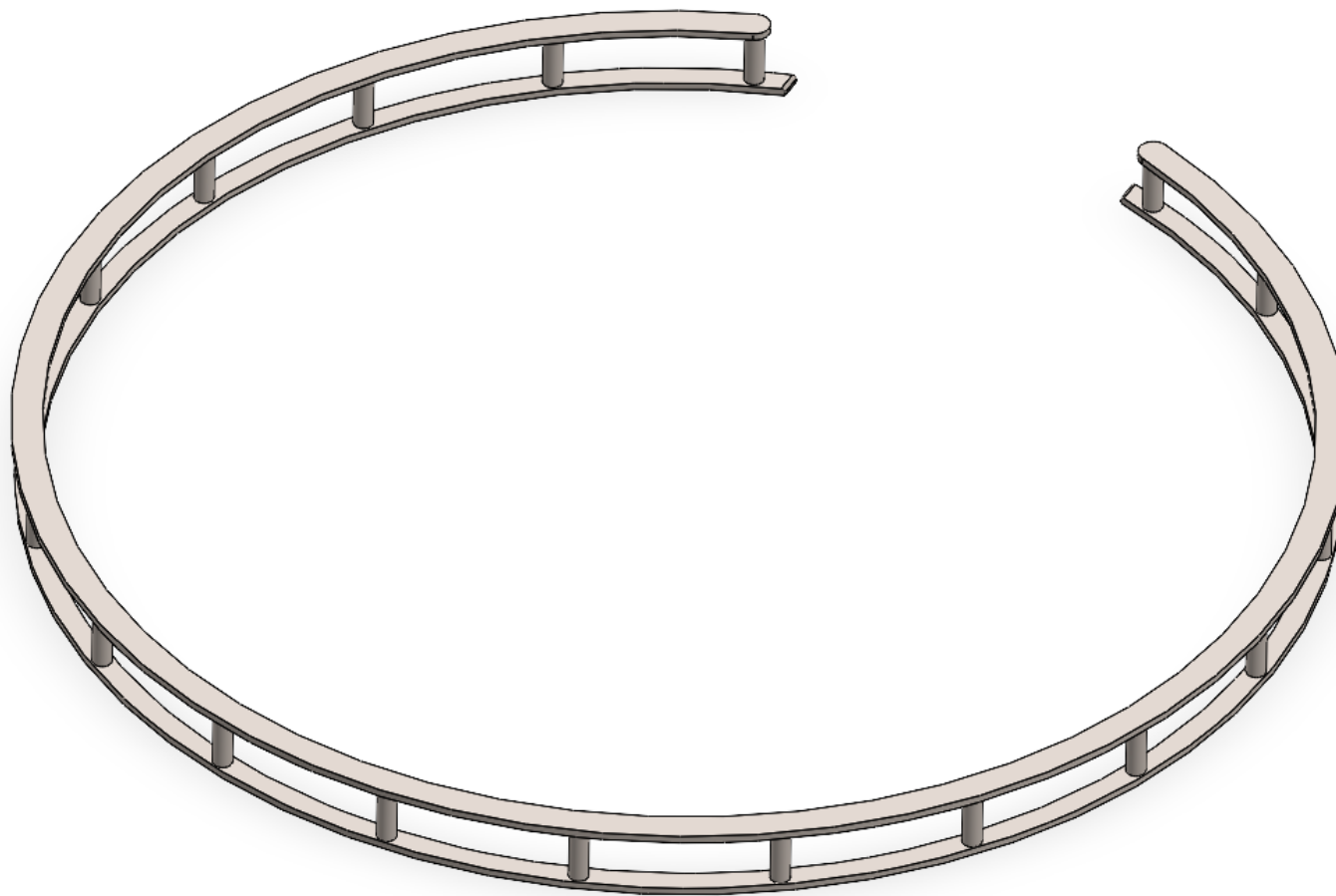


Material: Alloy Steel

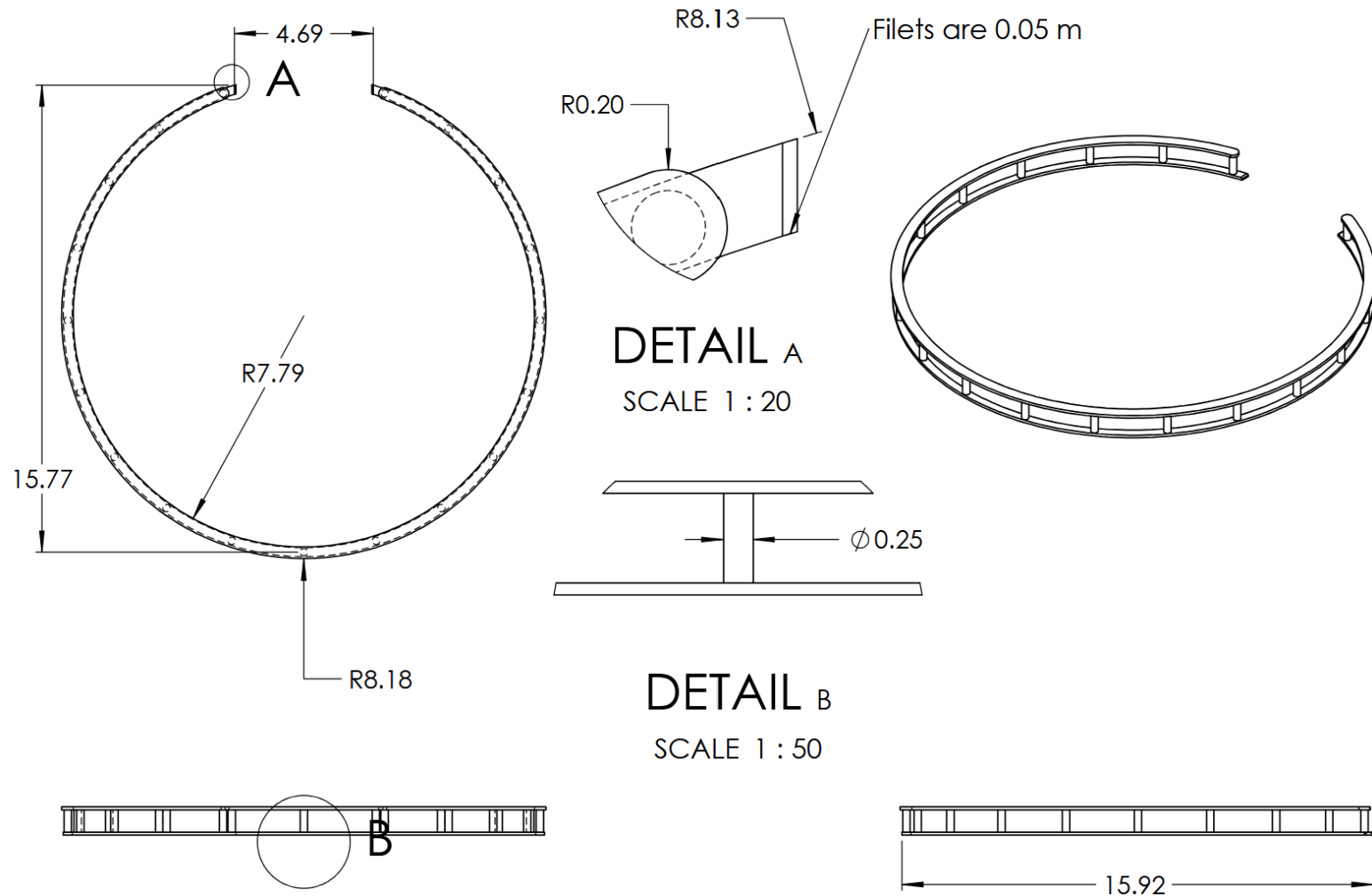
Mass: 53504.95 Kgs

Scale: 1:50	Title: Inside Stairs	ENME 272 Fall 2025
Units mks	Drawn by: Alicia Nguyen	Date: 12/8/2025

Inside Railing



Note:
19 bars total,
circular
patterned
160 degrees,
symmetric

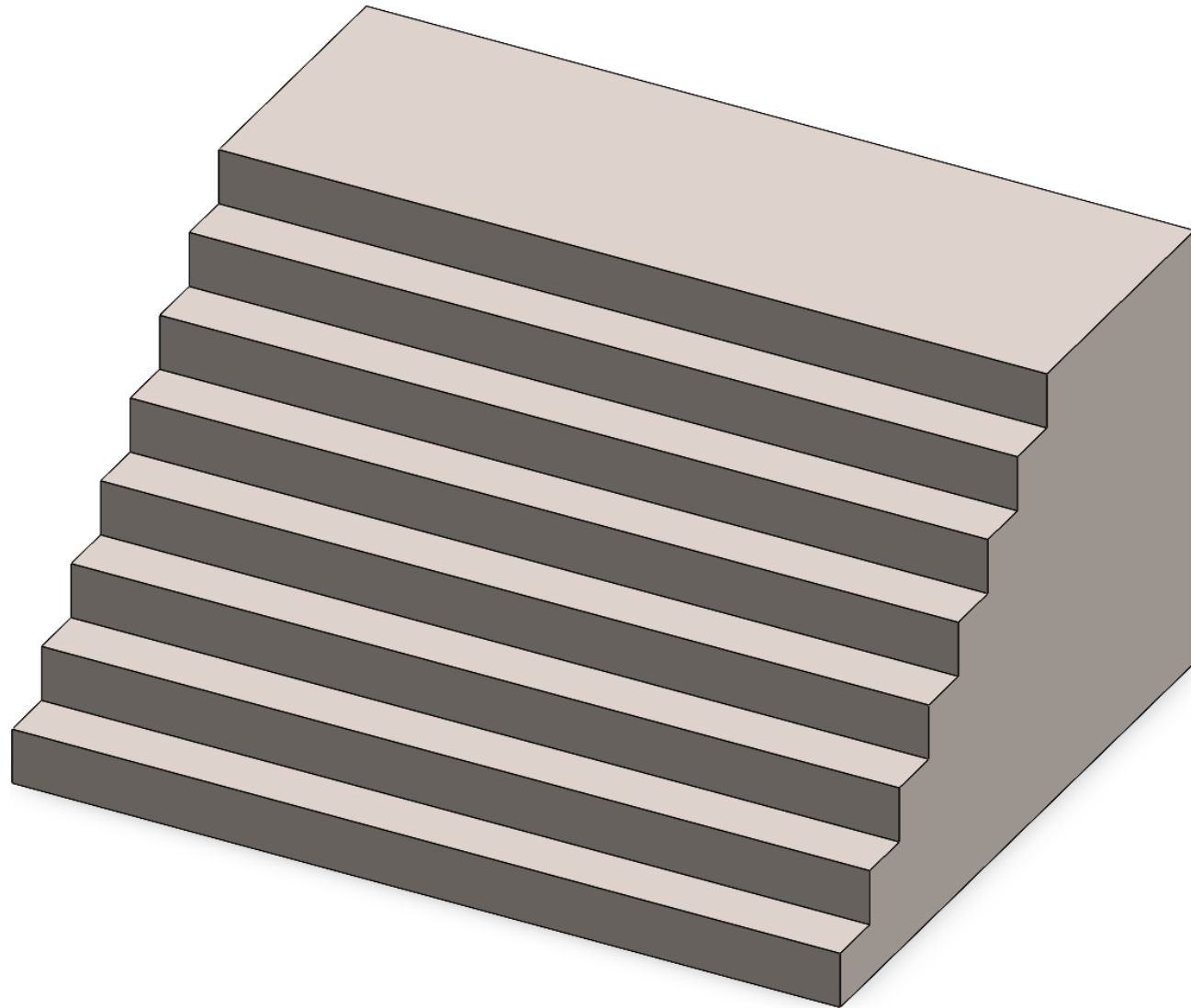


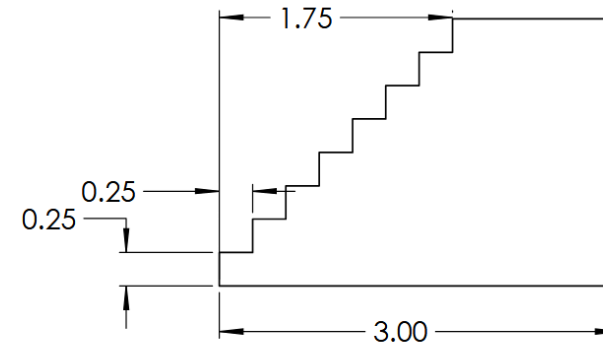
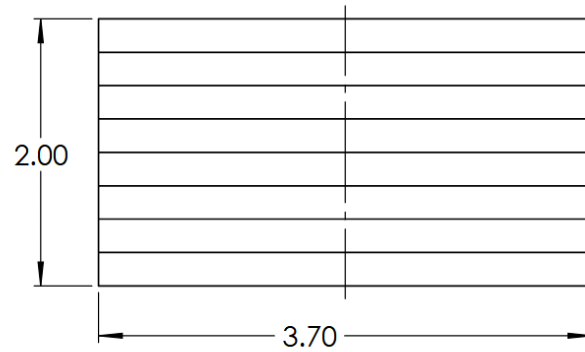
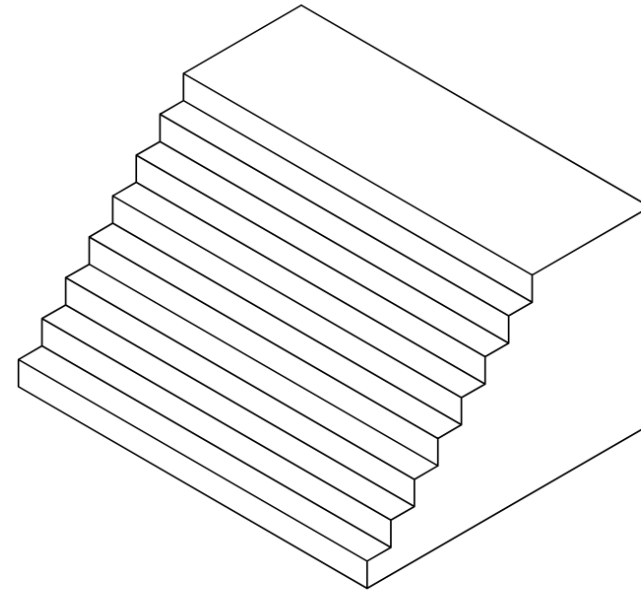
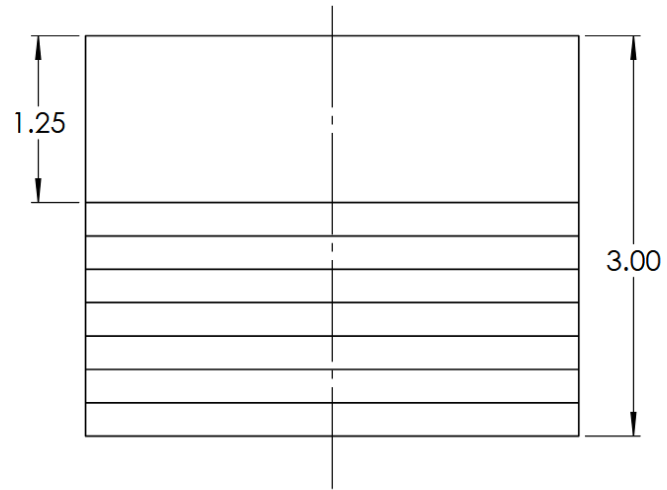
Material: Alloy Steel

Mass: 29434.18 Kg

Scale: 1:200	Title: Inside Railing	ENME 272 Fall 2025
Units mks	Drawn by: Alicia Nguyen	Date: 12/7/2025

Outside Stairs



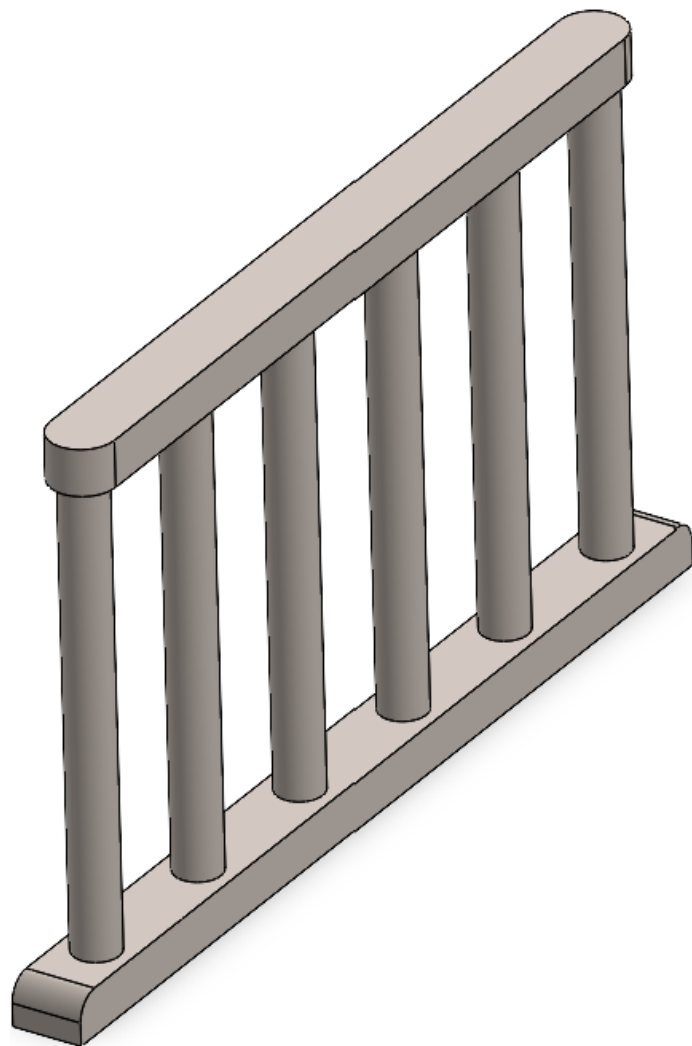


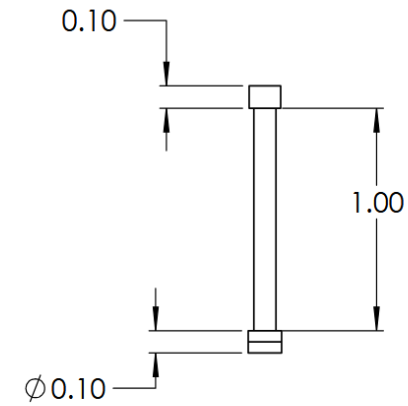
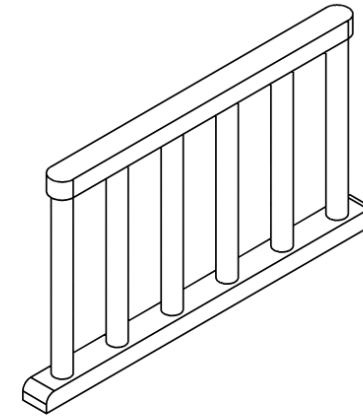
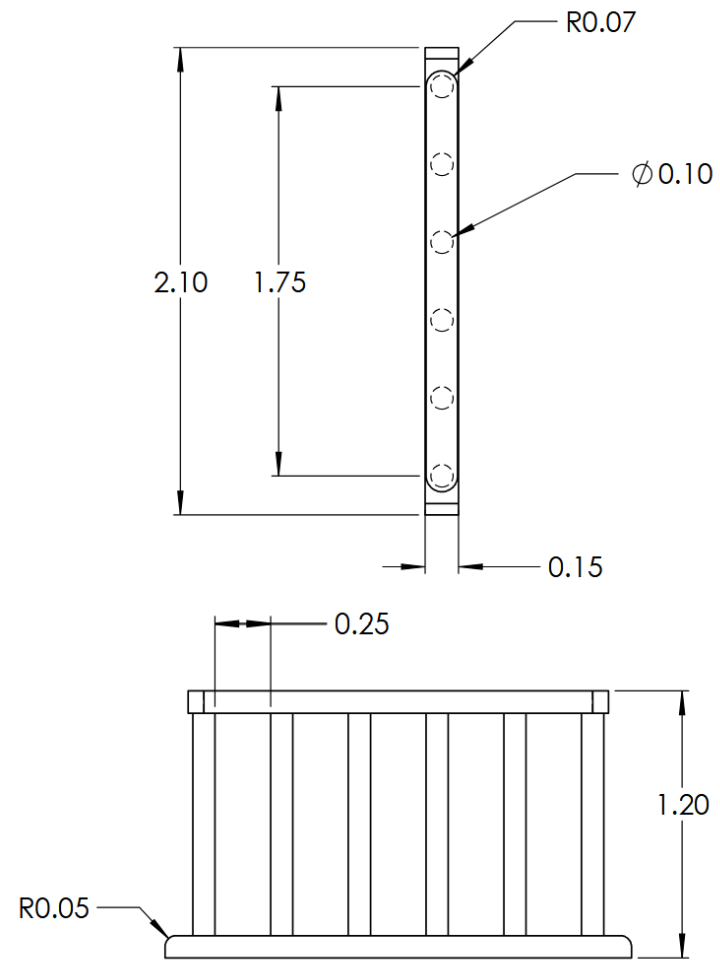
Material: Alloy Steel

Mass: 121082.50 Kgs

Scale: 1:50	Title: Outside Stairs	ENME 272 Fall 2025
Units mks	Drawn by: Alicia Nguyen	Date: 12/8/2025

Outside Railing



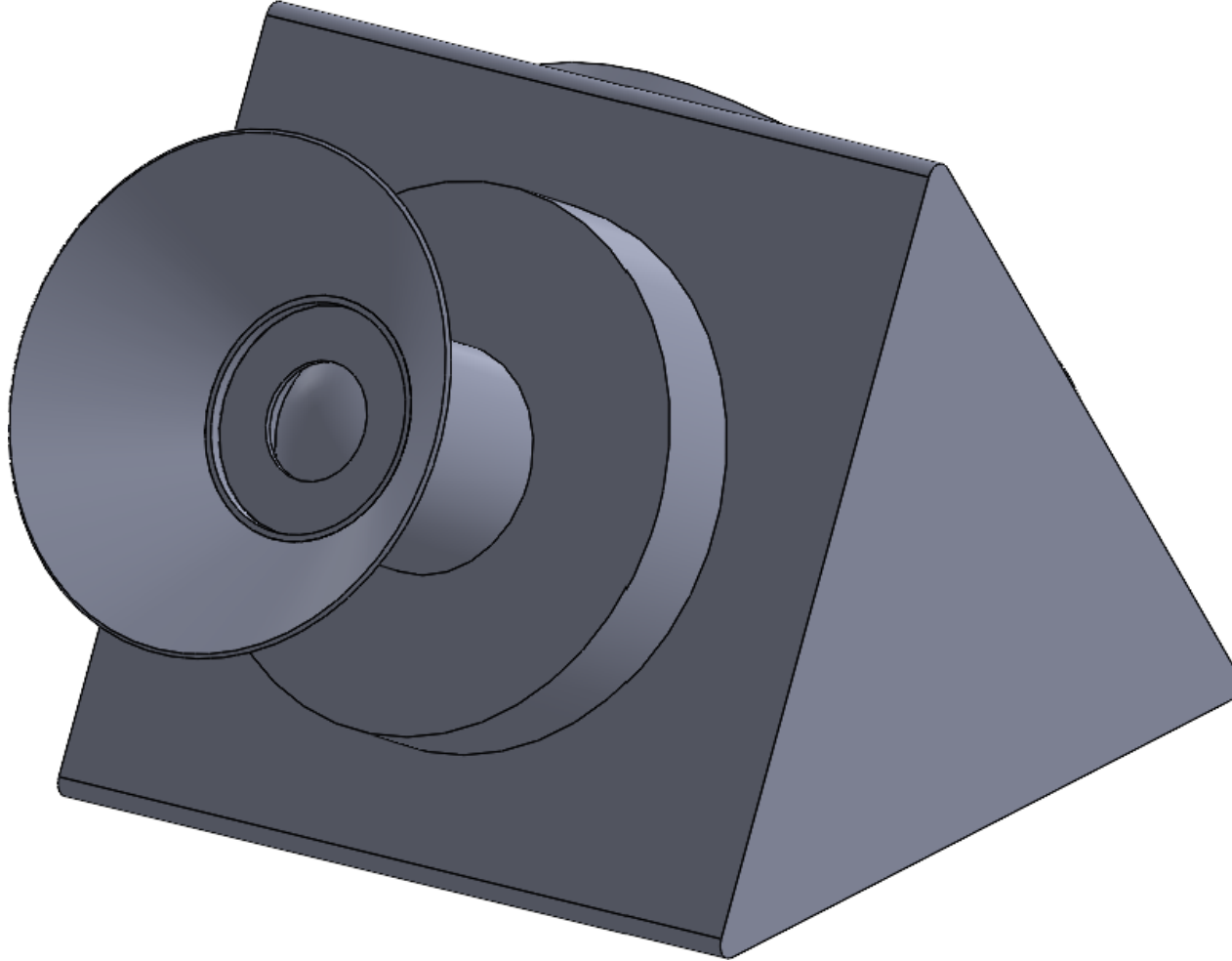


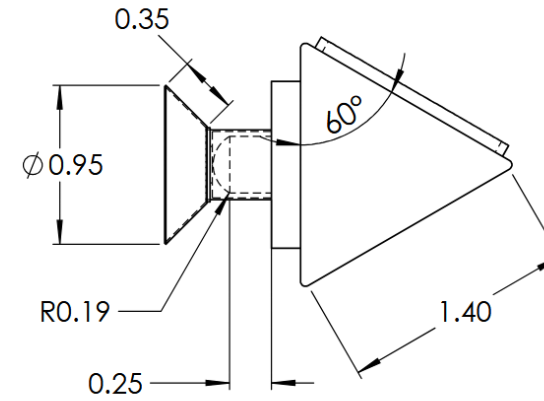
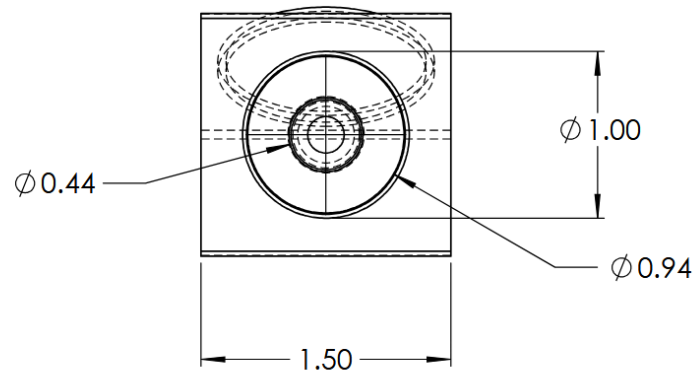
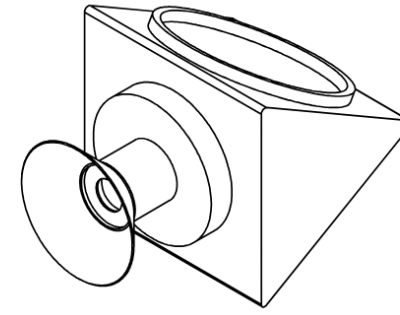
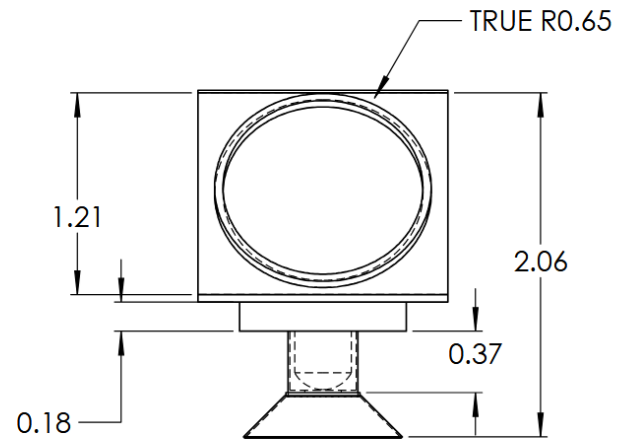
Material: Alloy Steel

Mass: 807.21 Kg

Scale: 1:30	Title: Outside Railing	ENME 272 Fall 2025
Units mks	Drawn by: Alicia Nguyen	Date: 12/7/2025

Eyeglass Piece



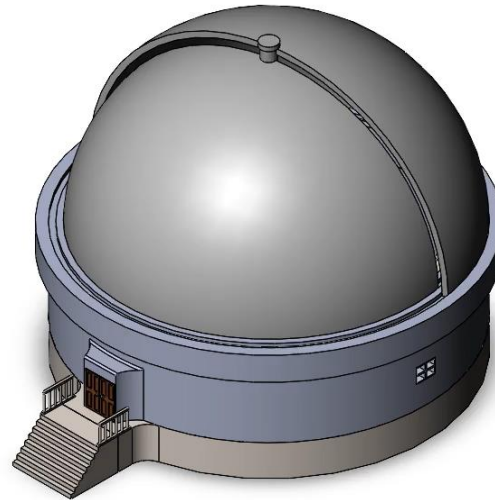


Material: 6061 Aluminum

Mass: 4461.46 Kg

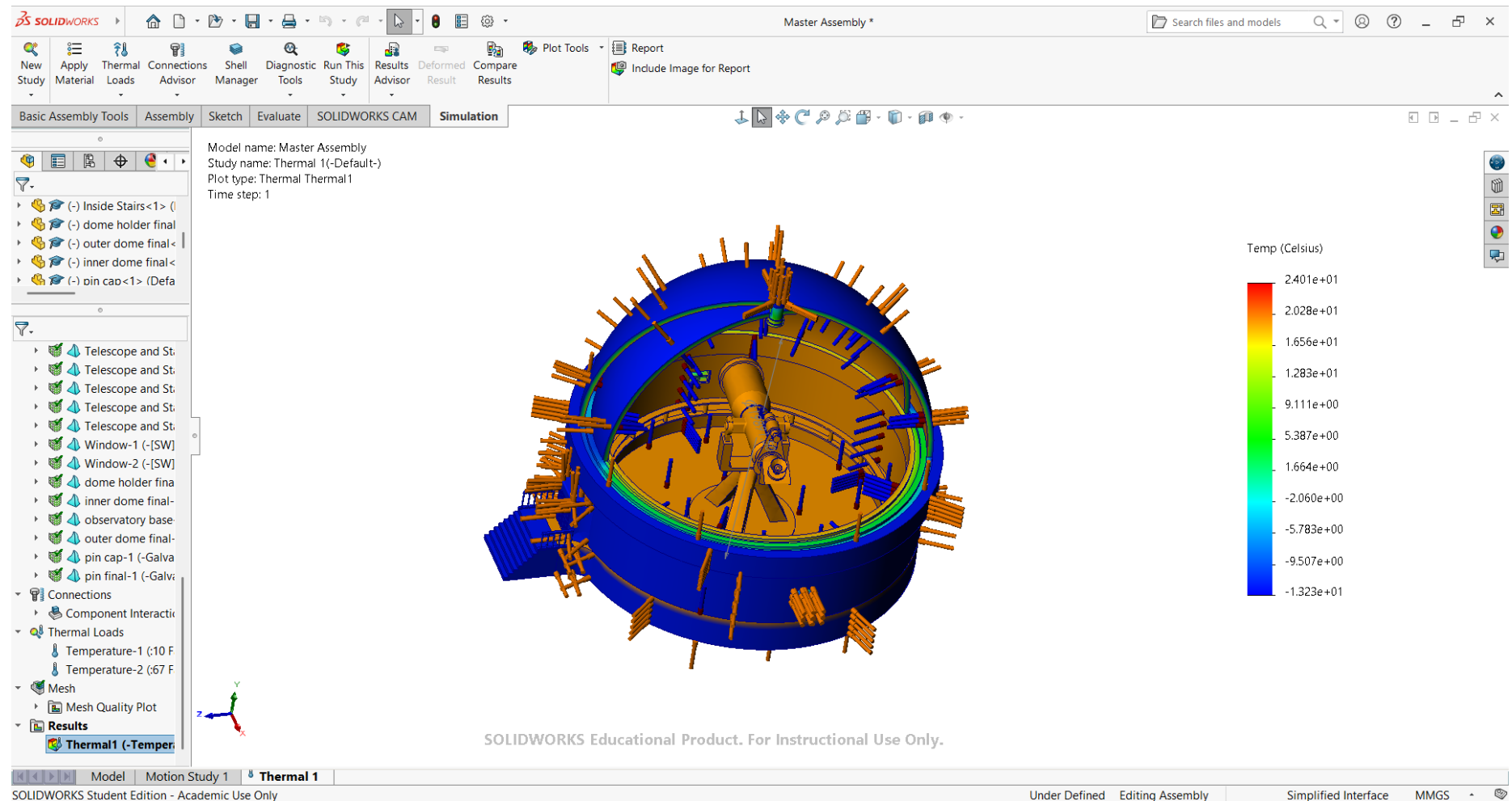
Scale: 1:40	Title: Watchglass	ENME 272 Fall 2025
Units mks	Drawn by: Alicia Nguyen	Date: 12/7/2025

Animation



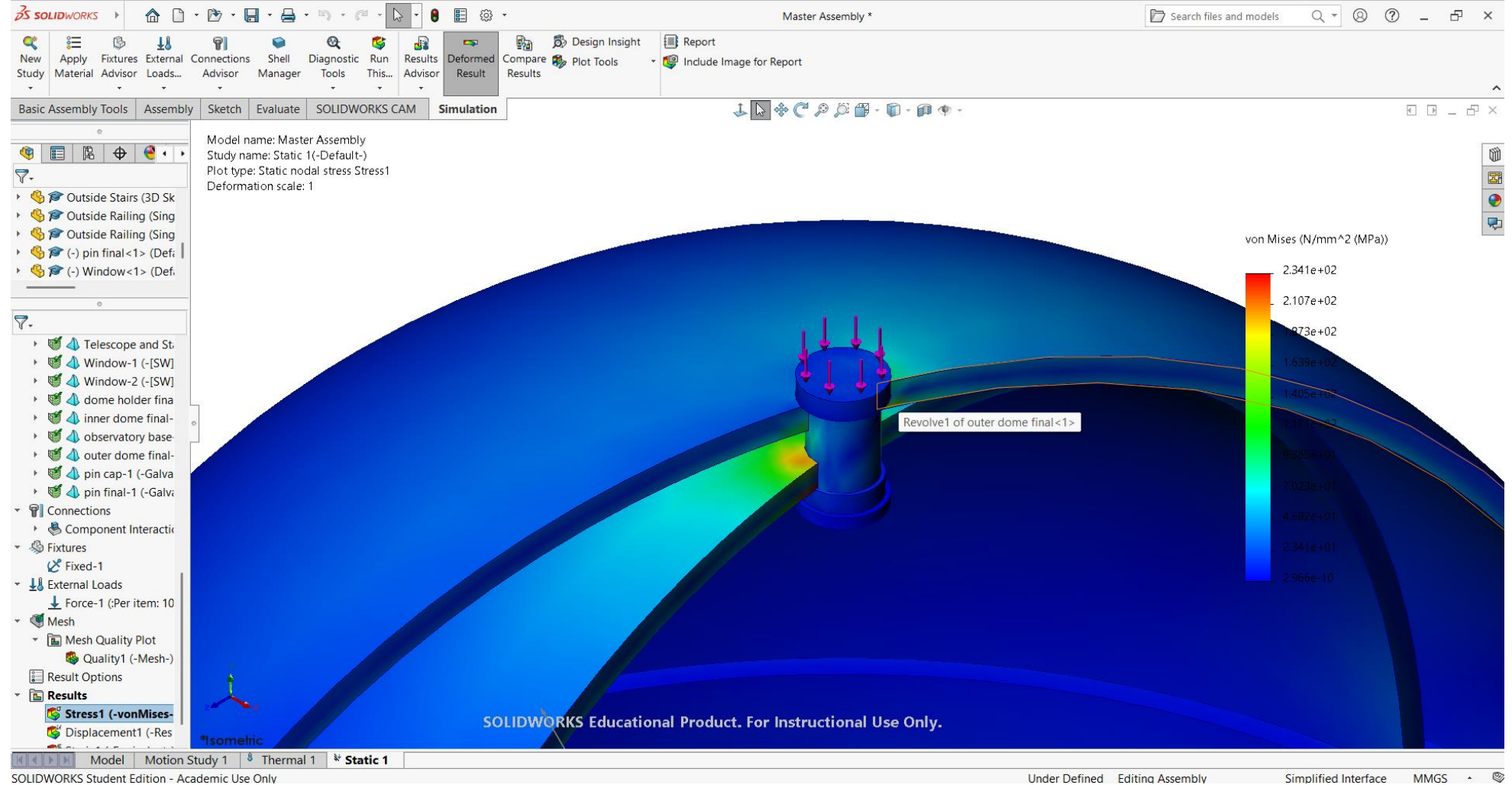
- Chose inner and outer dome plates and telescope base to show rotational motion
- Chose telescope and telescope sliders to show translational (vertical) motion
- Key takeaway: make a subassembly flexible first to move in the main assembly

Thermal FEA



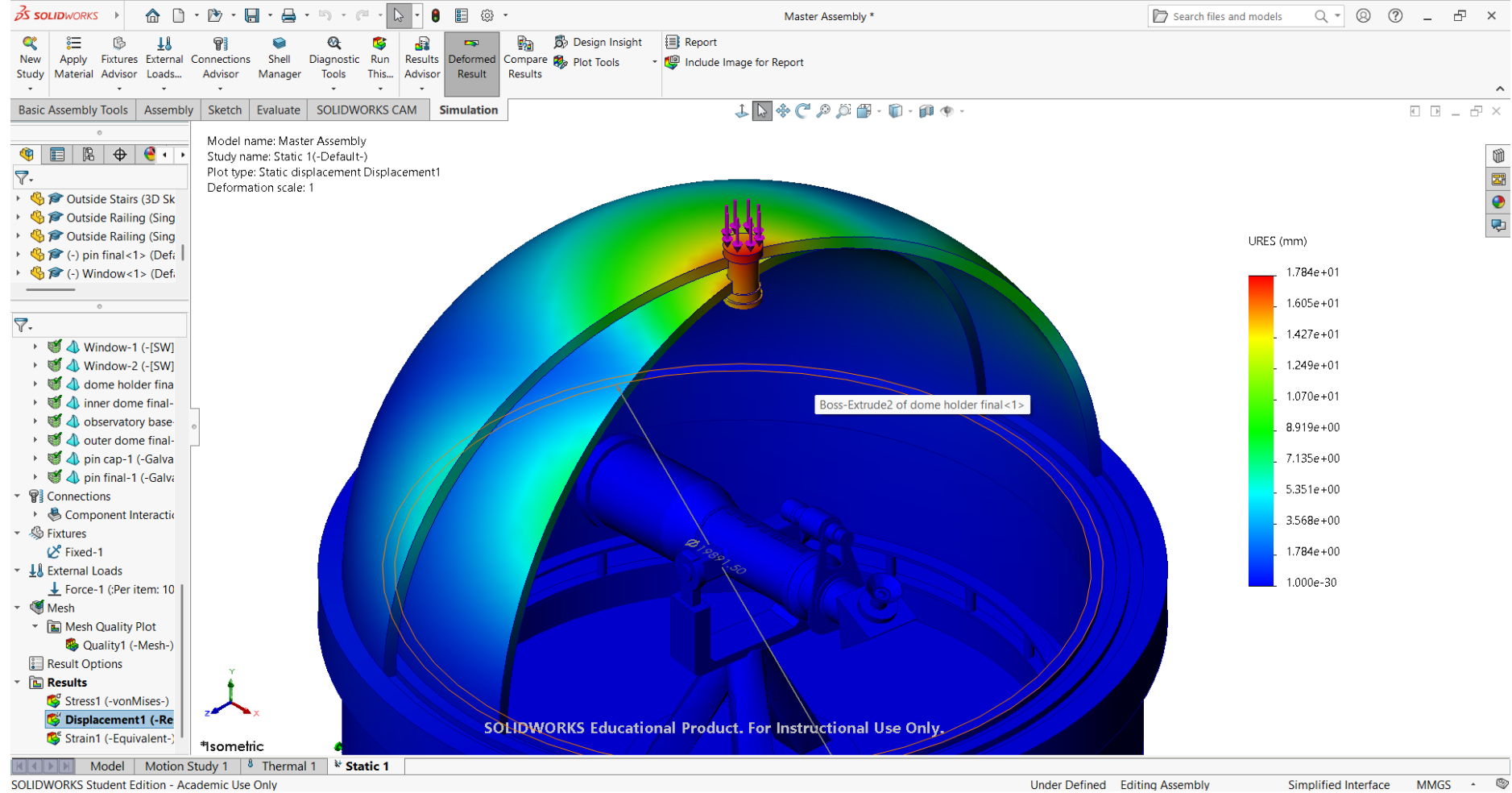
We wanted to know how well insulated this building is to see if the observatory would be able to keep the telescope in ideal conditions. We made the outside 10° F to represent a cold day in Maryland and the inside 67° F to represent a heated observatory. After running the FEA we see that the observatory is able to keep the telescope at 67° F or ~19° C. This shows that the observatory is able to keep the telescope at ideal conditions in Maryland. We could improve the wall thickness or material in order to insure that the observatory is insulated enough

Static FEA (Stress)



We did this static FEA in order to figure out what happens if a huge load is applied to our dome and how that affects the rest of the building. For this FEA we put a load of 10,000,000N on the top pin and fixed the base of the observatory. The only place that felt stress was at the top near the pin and the max stress it felt was 234.1 N/mm². We used galvanized steel which has a yield strength of 203.9 N/mm². This means that with a force of 10,000,000 N it will finally fail - which is completely unreasonable to expect from air pressure – meaning that the dome would not fail under normal loading.

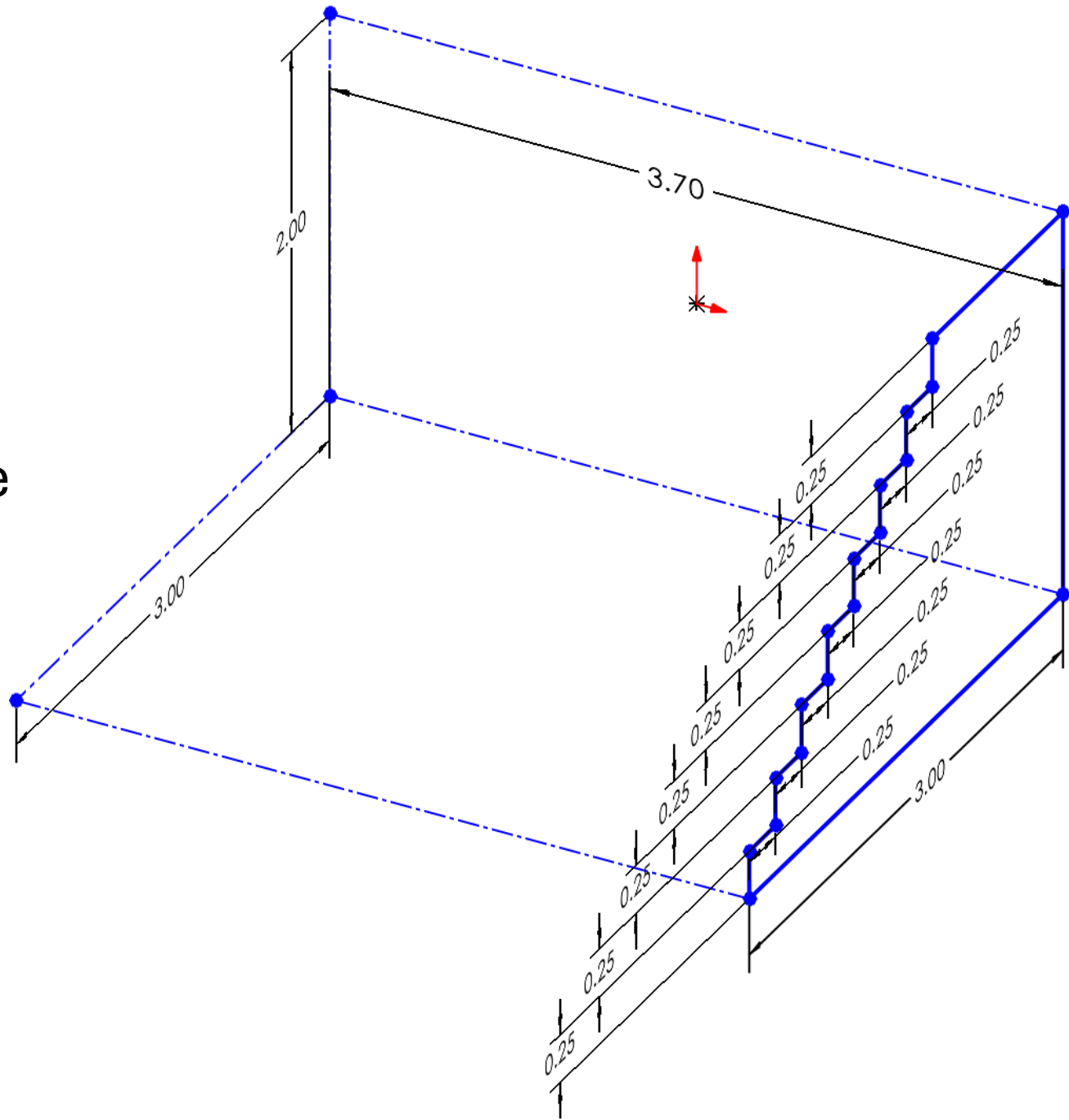
Static FEA (Displacement)



Our dome had a max displacement of 17.84mm on the pin and right next to the pin. Knowing this we could improve our design by adding supports to the bottom of the dome or get an even stronger material.

New feature

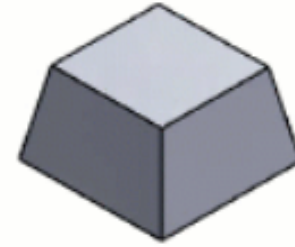
- 3d sketch
- Construction lines for viewing reference
- Helpful tool to see how your part will look before adding feature



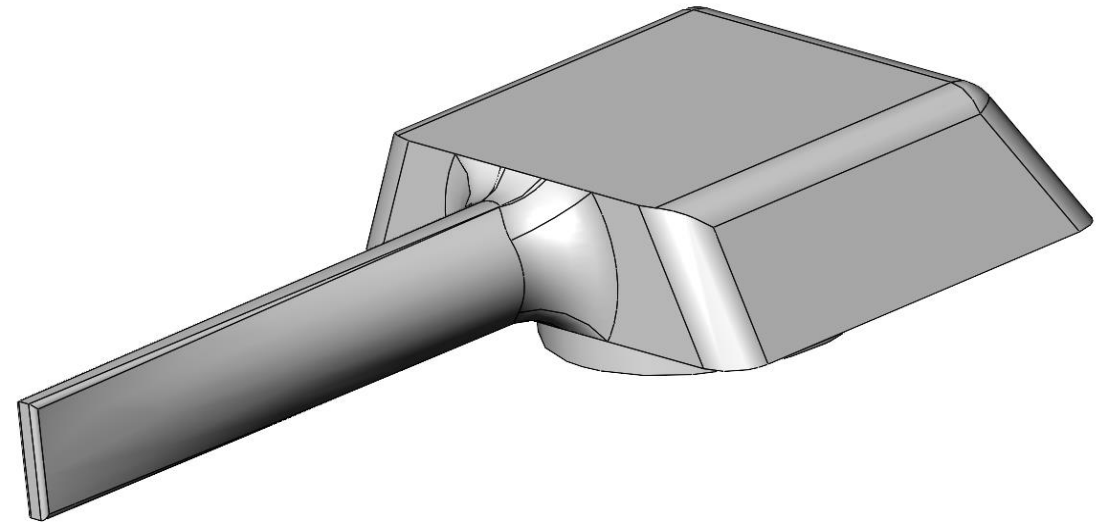
New feature

- Draft
- Tapers selected faces
- Helpful as a time saver, can also be used in the extrude menu

Draft



Tapers model faces by a specified angle, using a neutral plane or a parting line. Draft is often used to make a molded part easier to remove from the mold.



Reflections

Overall, the observatory was a really fun project that we all enjoyed working on.

- One of the problems that occurred most often was ensuring that everything was properly scaled before we assembled the parts in the final model.
- We spent a lot of time learning how to design the parts so they looked more realistic and like a proper telescope and observatory. Being able to look at reference photos was a huge help. Taking the extra steps to make our model look more visually appealing in SolidWorks was another learning curve; however, we believe this extra time taken paid off in the end.
- We also could have spent more time defining proper materials for our parts that mirror their real-world counterparts

References



- <https://grabcad.com>



<https://scmuseum.org>

References



<https://www.mtwilson.edu/>



<https://ciera.northwestern.edu>



Image found on google,
can't find link for website